

12

Aggregate Demand and Aggregate Supply

An expanded, lecture-style companion to the Chapter 12 slide deck — every slide restated, every mechanism spelled out, every figure reproduced, with exam traps, worked statements and practice questions.

36 slides covered

11 figures reproduced

Open questions + MCQ practice

Built to be read *alongside* the video lectures: the grey boxes reproduce what is on the slide, everything around them is the explanation that is given verbally.

Source deck: Macro_AY2526_Chapter12 — for use with Mankiw & Taylor, *Economics*, 6th edition (Cengage EMEA).

§0 How to use this guide

This document follows the Chapter 12 deck **slide by slide, in the original order**, so you can keep it open next to the video lecture. Each slide gets three things:

1. **What is literally on the slide** — reproduced in a blue-grey box, so you always know where you are in the deck and what the examinable "official" wording is.
2. **The expansion** — the reasoning, the intuition, the causal chain, the "why does that follow?" that a lecturer says out loud but does not type onto a slide.
3. **Exam handles** — tips, traps, and the specific way this point tends to be tested in open questions and MCQs.

The colour code

Box	What it contains
Blue-grey "On the slide"	Verbatim slide content. This is the examinable core. If you memorise only one thing per section, memorise this.
Green "Definition"	A term you must be able to define in one sentence in an open question.
Amber "Exam tip"	How this point is typically asked, and what a full-mark answer contains.
Red "Trap"	A mistake that is made every single year. Usually a movement-vs-shift confusion or a short-run/long-run confusion.
Purple "Going deeper"	Beyond the slide. Not required, but it makes the MCQs much easier because you understand <i>why</i> rather than <i>that</i> .
Blue "Link"	Connection to earlier chapters (IS-LM, the money market, the classical model) that the professor assumes you still remember.
Dashed "Mechanism"	A causal chain written as arrows. These are gold for open questions: reproduce the chain and you have the answer.

Exam tip — the single most valuable habit

Almost every question in this chapter can be answered with the same four-step routine: **(1)** Which curve moves — AD, SRAS, or LRAS? **(2)** Which direction? **(3)** What happens to P and Y in the **short run**? **(4)** What happens as expectations adjust, i.e. in the **long run**? Train yourself to write those four steps automatically, then add the mechanism in words. That structure alone typically earns most of the marks.

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§1 Three key facts about economic fluctuations

SLIDE 2 — ON THE SLIDE

Three Key Facts about Economic Fluctuations

- Economic fluctuations are irregular and unpredictable.
 - Fluctuations in the economy are often called the business cycle.
- Most macroeconomic variables fluctuate together.
 - Most macroeconomic variables that measure some type of income or production fluctuate closely together.
 - Although many macroeconomic variables fluctuate together, they fluctuate by different amounts.
- As output falls, unemployment rises.
 - Changes in real GDP are inversely related to changes in the unemployment rate.
 - During times of recession, unemployment rises substantially.

This slide is the *motivation* for the whole chapter. Before building a model, you first need to know what the model has to explain. These three facts are the empirical regularities that any theory of the short run must reproduce — and the AD-AS model is built precisely so that it does.

Fact 1 — Fluctuations are irregular and unpredictable

Real GDP does not grow smoothly. It grows on average, but around that trend it wanders: some years are strong, some years are weak, and occasionally output actually falls. Those episodes are what we call **recessions**.

Definitions you should be able to write

Business cycle — the irregular, recurring fluctuations of real economic activity (output, employment, incomes) around its long-run trend.

Recession — a period of falling real income and rising unemployment. A widely used rule of thumb is two consecutive quarters of falling real GDP, though official dating bodies look at a broader set of indicators.

Depression — a severe and prolonged recession.

The word "cycle" is genuinely misleading, and that is the point the slide is making with the word **irregular**. A cycle in physics has a fixed period and a fixed amplitude: if you know where you are, you know when the next peak arrives. The business "cycle" has neither. Expansions have lasted anywhere from a couple of years to more than a decade; recessions have been shallow and short or deep and long. There is no reliable calendar.

Unpredictable is a second, stronger claim. Forecasters have a poor record of calling turning points in advance. Economists therefore talk about *fluctuations*, and they model them as the economy being hit by **shocks** — unexpected disturbances to demand or to supply. That framing

is exactly how the rest of the chapter proceeds: we take an economy sitting in long-run equilibrium, hit it with a shock, and trace out what happens first and what happens eventually.

Exam tip

If asked "why is the term 'business cycle' somewhat misleading?", the answer is: it suggests regularity and predictability (fixed period, fixed amplitude) whereas real fluctuations are irregular in both length and depth and cannot be reliably forecast.

Fact 2 — Most macroeconomic variables fluctuate together

When real GDP falls, it is almost never one isolated variable misbehaving. Consumption, investment, employment, hours worked, corporate profits, retail sales and tax revenues all tend to move down together, and up together in an expansion. The technical word is **co-movement**, and variables that move in the same direction as GDP are called **procyclical** (variables that move in the opposite direction, such as unemployment, are **countercyclical**).

Why does this matter for the model? Because it justifies the enormous simplification we are about to make. If the components of GDP all moved independently, we would need a separate story for each one. Since they move together, we can summarise the state of the economy with a **single number for output** (real GDP, on the horizontal axis) and a **single number for prices** (the price level, on the vertical axis). The whole AD-AS diagram rests on this fact.

The second sub-bullet — "**they fluctuate by different amounts**" — is the part students skip and examiners like. Co-movement is about *direction*; amplitude differs a lot across components:

Typical cyclical behaviour of the components of GDP

Component	Volatility over the cycle	Why
Investment (I)	Very high	Investment is <i>postponable</i> . A firm can simply wait a year before building a new plant, and a household can delay buying a house. It is also highly sensitive to interest rates and to expectations about the future, both of which swing sharply in a downturn.
Consumer durables	High	Cars, furniture, appliances — same logic as investment: the old one can last another year.
Consumption of non-durables and services (C)	Low	Food, rent, heating, healthcare cannot be postponed. Households also smooth consumption: they save in good years and dis-save in bad ones, so spending moves less than income.
Government purchases (G)	Low / policy-driven	Set by political decision, not by the cycle. It may even move counter-cyclically if governments deliberately stimulate in a downturn.

Exam tip

"Investment is the most volatile component of GDP over the business cycle" is a standard true/false or MCQ item. The reason to give is **postponability + interest-rate sensitivity +**

sensitivity to expectations/confidence. This also foreshadows §5: a change in business confidence shifts AD, and it shifts it by a lot.

Fact 3 — As output falls, unemployment rises

This is the **inverse relationship** between real GDP growth and the change in the unemployment rate, known in the literature as **Okun's law**. The logic is mechanical: firms produce goods and services using workers, so when firms cut production they need fewer worker-hours. Some of the adjustment happens through hours and overtime, but a substantial part happens through employment — hiring freezes, non-renewal of temporary contracts, and layoffs. Unemployment therefore rises substantially in recessions.

Two refinements that are worth a sentence in an open question:

- **The relationship is between *changes*, not levels.** It is not "low GDP means high unemployment" in some absolute sense; it is that a *fall* in real GDP relative to trend goes together with a *rise* in the unemployment rate.
- **Employment falls by less than output, at least initially** — firms engage in **labour hoarding**: they keep skilled workers they have invested in training, because firing and re-hiring is expensive. This is why productivity often dips at the start of a recession.

Link — why fact 3 is structural for this chapter

Fact 3 is not just an observation; it is the hinge that lets us translate between the **goods market** (output Y) and the **labour market** (unemployment u). You will need it twice later: in §7, where the mechanism behind the upward-sloping SRAS runs *output* → *employment* → *unemployment* → *wages* → *prices*; and in §10, where the natural rate of output Y_n is defined as the output level at which unemployment equals its natural rate u_n . Keep the mapping in your head as: higher $Y \Leftrightarrow$ lower u .

§2 Why the short run is different from the long run

SLIDE 3 — ON THE SLIDE

Explaining Short-Run Economic Fluctuations

- How the Short Run Differs from the Long Run
 - Most economists believe that classical theory describes the world in the long run but not in the short run.
 - Changes in the money supply affect nominal variables but not real variables in the long run.
 - The assumption of monetary neutrality is not appropriate when studying year-to-year changes in the economy.

This single slide is the intellectual pivot of the entire course. Everything you learned in the "long-run" chapters is now going to be declared *true, but only eventually*.

The classical world you are leaving behind

Definitions

Classical dichotomy — the theoretical separation of variables into *real* variables (quantities and relative prices: real GDP, employment, the real wage W/P , the real interest rate) and *nominal* variables (measured in money units: the price level, the nominal wage, the money supply).

Monetary neutrality — the proposition that changes in the money supply affect only nominal variables and leave all real variables unchanged.

In the classical model, real GDP is determined entirely on the **supply side**: by the available capital, the available labour, the natural resources, and the technology that turns them into output. Money is a "veil". Double the money supply and, in the long run, every price and every nominal wage doubles; nothing real changes; you have simply changed the units in which value is measured. Nobody is richer or poorer, nobody produces more, nobody works more.

Why that cannot be the whole story

The classical picture predicts that a monetary contraction should have no effect on production or employment. Empirically that is plainly false *in the short run*: monetary tightening is followed by recessions, and demand collapses are followed by mass unemployment, not by an instant, painless fall in all prices and wages. Fact 1 and Fact 3 from slide 2 are exactly the facts a purely classical model cannot generate.

The reason the classical result fails in the short run is that prices and — above all — nominal wages do not adjust instantaneously. They are set in advance, in contracts, on the basis of what people *expected* the price level to be. When something unexpected happens, quantities move before prices have finished moving. That is the entire content of the phrase "the assumption of monetary neutrality is not appropriate when studying year-to-year changes in the economy".

The two regimes side by side — memorise this table

	Short run (this chapter's focus)	Long run (classical chapters)
Prices and nominal wages	Sticky / slow to adjust; expectations of prices are backward-looking and lag behind	Fully flexible; expectations have caught up ($P^e = P$)
Monetary neutrality	Does not hold — money and demand affect real output	Holds — money affects only nominal variables
What determines output	Both demand and supply: the intersection of AD and SRAS	Supply-side factors only: labour, capital, natural resources, technology
Aggregate supply curve	Upward sloping (SRAS)	Vertical at Y_n (LRAS)
Effect of a demand shock on Y	Yes, output moves	No, only the price level moves

Trap

Students often write "the classical model is wrong". That is not the claim and it will cost you marks. The claim is that the classical model is **the correct description of the long run**, and that the AD-AS model of this chapter is a description of how the economy *travels towards* that long run. In this chapter the classical result reappears at the end of every experiment: after a demand shock, once expectations adjust, output is back at its natural rate and only the price level has permanently changed. Money is neutral again — just not immediately.

Exam tip

A very common open question is: "Explain the difference between the short run and the long run in macroeconomics." A full-mark answer contains three elements: (i) the definition is about **price and wage flexibility**, not about a fixed number of months; (ii) in the long run the classical dichotomy and monetary neutrality hold, in the short run they do not; (iii) the operational consequence — LRAS is vertical, SRAS is upward sloping, and demand shocks move output only in the short run.

§3 The basic model: from IS-LM to AD-AS

SLIDE 4 — ON THE SLIDE

The Basic Model of Economic Fluctuations

- IS-LM model
 - Does not account for changes in the price level
 - Focus on aggregate demand
- The Model of Aggregate Demand and Aggregate Supply
 - Used to explain short-run fluctuations in economic activity around its long-run trend.
 - Focus on two variables:
 - The economy's output of goods and services measured by real GDP.
 - The overall price level measured by the CPI or the GDP deflator.

What was missing from IS-LM

Recall what the IS-LM model does. It has two markets:

- the **goods market**, summarised by the *IS* curve — combinations of the interest rate i and income Y at which planned expenditure equals output;
- the **money market**, summarised by the *LM* curve — combinations of i and Y at which the demand for real money balances equals the supply M/P .

Their intersection gives equilibrium Y and i . But notice what is being held fixed: **the price level P** . It appears in the model only inside the term M/P , and it is treated as an exogenous constant. IS-LM therefore answers the question "how much output is demanded, *given* the price level?" — it is a theory of **aggregate demand only**. It has literally nothing to say about inflation, about supply shocks, or about how the economy returns to its natural rate.

Link — the two limitations, stated the way the professor states them

(1) "Does not account for changes in the price level." P is exogenous, so the model cannot explain what happens to prices, and cannot handle any shock that works through prices or costs.

(2) "Focus on aggregate demand." The whole of IS-LM sits on the demand side. There is no production function, no labour market, no wage bargaining — nothing that tells you whether firms are *willing* to supply the output that is being demanded.

AD-AS fixes both: it makes P endogenous, and it adds a supply side.

The new model in one sentence

AD-AS is a model with **two endogenous variables** — real GDP Y and the price level P — and **two curves** in (Y, P) space whose intersection determines them jointly. IS-LM is not discarded: it is *absorbed*, and it becomes the AD curve.

Exam tip — the measurement footnote is examinable

The slide says the price level is "measured by the CPI or the GDP deflator". Know the difference: the **CPI** measures the price of a fixed basket bought by a typical consumer and includes imported goods; the **GDP deflator** measures the price of all goods and services *produced domestically* and its basket changes automatically each year. For the purposes of the diagram they are interchangeable — both are indices of the overall price level.

SLIDE 5 — ON THE SLIDE

The Basic Model of Aggregate Demand and Aggregate Supply

- The **aggregate-demand curve** shows the quantity of goods and services that households, firms, and the government want to buy at each price level.
 - Derived from the IS-LM model
- The **aggregate-supply curve** shows the quantity of goods and services that firms choose to produce and sell at each price level.
 - Derived from the wage- and price-setting behaviour of workers and firms

Read the two sub-bullets carefully, because they tell you exactly where each curve comes from in *this* course, and they are the difference between the professor's version and the plain textbook version:

Curve	Derived from	What that means in practice
AD	The IS-LM model	For each value of P , solve IS-LM for equilibrium Y . Plot the pairs. The AD curve is the <i>locus of IS-LM equilibria</i> as the price level varies. Consequence: anything that shifts IS or LM (other than P itself) shifts AD.
AS	The wage- and price-setting behaviour of workers and firms	The supply side is a <i>labour-market</i> story: unions and employers bargain a nominal wage; firms then set prices as a mark-up over their costs. Consequence: anything that changes bargained wages or firms' costs shifts AS.

Trap — the most important conceptual trap in the chapter

The AD curve looks like a microeconomic demand curve, but it slopes down for **completely different reasons**. A micro demand curve for pizza slopes down because when pizza gets cheaper you substitute *away from other goods* towards pizza, and because your real income rises. Neither applies here: along the AD curve **all prices move together**, so there is no "other good" to substitute towards, and there is no outside income to become worth more. The downward slope of AD must come from something else entirely — that is the whole point of slides 9–12. Never justify the AD slope with a substitution argument.

Figure 1 Aggregate Demand and Aggregate Supply.

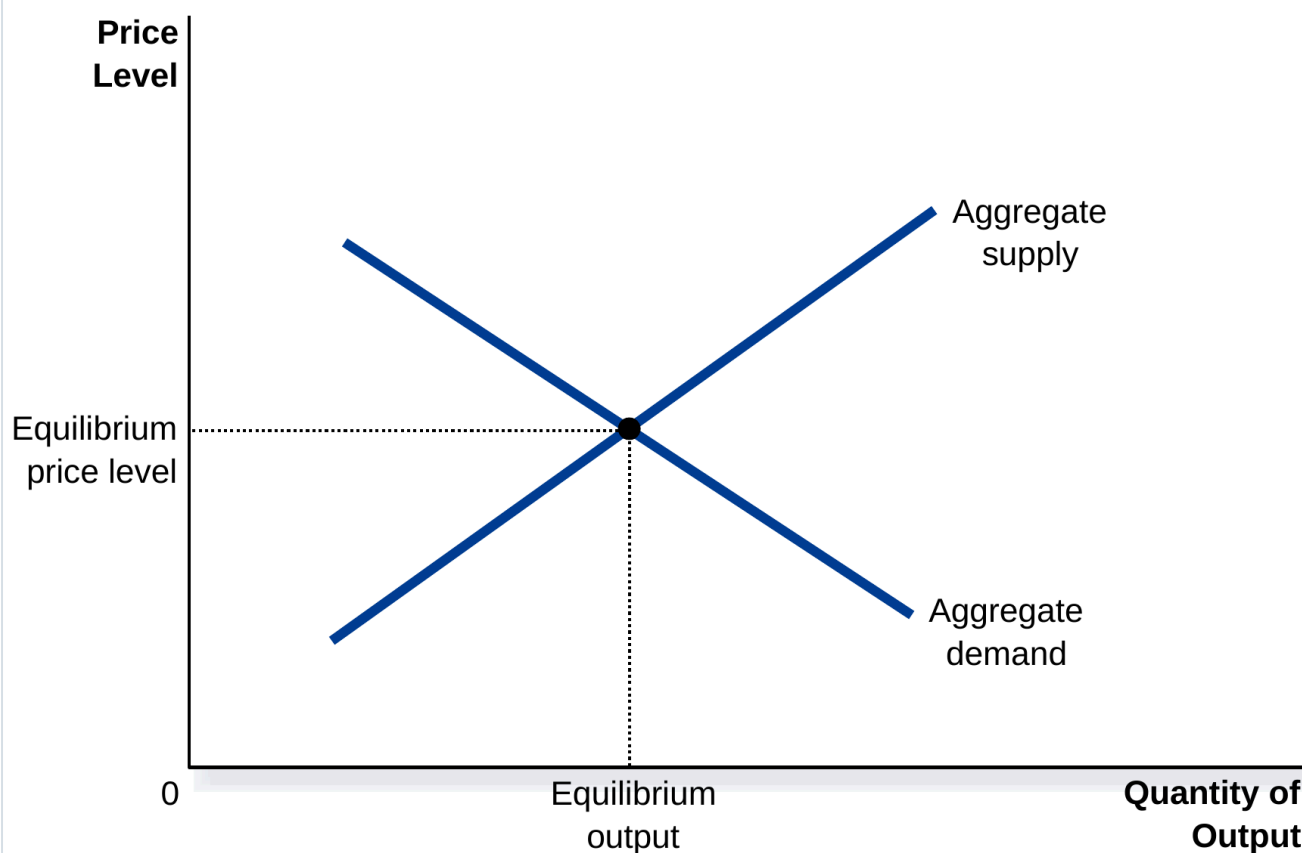


Figure 1 (slide 6) — Aggregate demand and aggregate supply. The axes are the two variables of the model: the **price level** (an index — *not* the price of any single good) on the vertical axis, and the **quantity of output** (real GDP) on the horizontal axis. The downward-sloping curve is aggregate demand, the upward-sloping curve is (short-run) aggregate supply. Their intersection determines the **equilibrium price level** and **equilibrium output** simultaneously. This is the picture the rest of the chapter manipulates: every experiment consists of moving one of these two curves and reading off the new intersection.

Exam tip — always label the axes correctly

Vertical axis: **Price level** (P) — not "price", not "inflation". Horizontal axis: **Quantity of output** / real GDP (Y) — not "quantity". Diagrams with sloppy axis labels lose easy marks, and confusing the price *level* with the *inflation rate* (the growth rate of the price level) is a genuine conceptual error, not just a labelling one.

§4 The aggregate demand curve and why it slopes downward

SLIDE 7 — ON THE SLIDE

The Aggregate Demand Curve

- The four components of GDP (Y) contribute to the aggregate demand for goods and services.
- $Y = C + I + G + NX$

The national income accounting identity is being reused here as a **checklist**. Total demand for domestically produced output has exactly four sources, so there are exactly four channels through which anything — including the price level — can affect aggregate demand:

Component	Who is spending	Does it respond to a change in the price level P ?
C	Households, on consumption	Yes — through the <i>wealth effect</i> (slide 10).
I	Firms and households, on capital goods and housing	Yes — through the <i>interest-rate effect</i> (slide 11). This is the main channel in this course.
G	Government, on goods and services	No — G is set by political decision, not by the price level. It is a shift variable, never a movement-along variable.
NX	Foreigners, net of our imports	Yes — through the <i>exchange-rate effect</i> (bracketed on slide 9, so secondary here).

So when the slide asks "why does AD slope downward?", the answer must be assembled component by component: a fall in P must raise C , or I , or NX , or some combination. It cannot raise G .

Figure 2 The Aggregate Demand Curve...

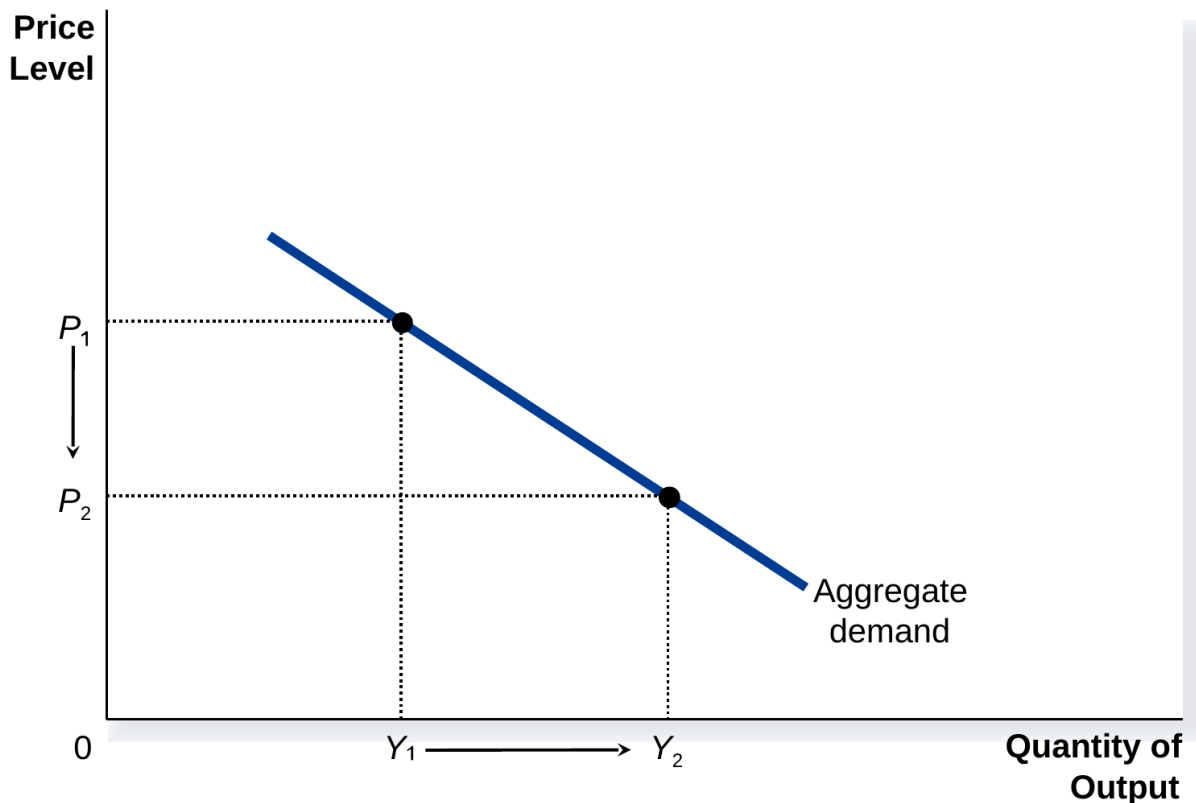


Figure 2 (slide 8) — The aggregate demand curve. A fall in the price level from P_1 to P_2 raises the quantity of goods and services demanded from Y_1 to Y_2 . Note carefully what this is: a **movement along** a given AD curve, caused by a change in the price level and nothing else. Everything else — confidence, taxes, government spending, the money supply, foreign income — is being held constant. This is the picture that the three effects on slide 9 have to justify.

SLIDE 9 — ON THE SLIDE

Why the Aggregate Demand Curve Is Downward Sloping

- The Price Level and Consumption: The Wealth Effect
- The Price Level and Investment: The Interest Rate Effect
- (The exchange rate effect)

Exam tip — read the brackets

The professor puts **the exchange-rate effect in parentheses**. In this deck, brackets consistently mean: "this exists, know that it exists and roughly what it says, but it is not the centre of gravity of the exam." The two unbracketed effects — wealth and interest rate — are the ones you must be able to explain in full, mechanism and all. The interest-rate effect is the one that is explicitly *derived* with a diagram (slide 12), which tells you it is the most important of the three.

Effect 1 — The wealth effect (consumption)

SLIDE 10 — ON THE SLIDE

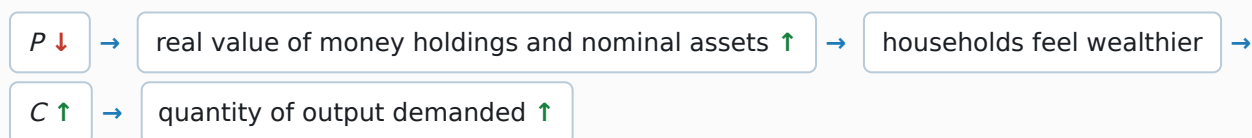
The Price Level and Consumption: The Wealth Effect

- A decrease in the price level makes consumers feel more wealthy, which in turn encourages them to spend more.
- This increase in consumer spending means larger quantities of goods and services demanded.

The missing step — the one that is said out loud and not written — is **why** a lower price level makes consumers wealthier. It is not a psychological trick. It is arithmetic about the **real value of nominally denominated assets**.

Households hold part of their wealth in assets whose value is fixed in money terms: cash, current-account and savings balances, bonds with a fixed nominal face value. The purchasing power of those assets is **nominal value ÷ price level**. If the price level falls, the same €10,000 in the bank buys more goods. Real wealth has risen even though nothing nominal changed. Richer households consume more — hence C rises and the quantity of output demanded rises.

MECHANISM — THE WEALTH EFFECT



Going deeper — why economists think this effect is small

Two qualifications that will make an open answer look genuinely well-informed. First, money holdings are a small share of total household wealth; most wealth is in housing, equity and pension claims, which are *real* or at least not fixed in nominal terms, so their real value is not mechanically raised by a fall in P . Second, and more fundamentally, for every holder of a nominally fixed asset there is an **issuer of a nominally fixed liability**. A fall in the price level raises the real value of savers' bank deposits, but it also raises the real burden of borrowers' mortgages. The gains and losses largely cancel across the economy, and if debtors have a higher propensity to consume than creditors, the net effect on aggregate consumption can even be negative. This is why the textbook treats the wealth effect as the **least important** of the three reasons, and why the professor derives the interest-rate effect rather than this one.

Effect 2 — The interest-rate effect (investment)

SLIDE 11 — ON THE SLIDE

The Price Level and Investment: The Interest Rate Effect

- Derived from the IS-LM model
- An increase in the price level decreases the quantity of real money balances
- A decrease in the quantity of real money balances increases the interest rate on the money market
- An increase in the interest rate decreases investment (I) and decreases aggregate demand

This is the core of the chapter's demand side, and the slide already gives you the chain in four links. Here is what sits behind each link.

Link 1: $P \uparrow \Rightarrow M/P \downarrow$. The central bank controls the **nominal** money supply M . What matters for transactions is the **real** money supply M/P — the purchasing power of the money in circulation. Hold M fixed and raise P : real money balances fall mechanically. This is pure arithmetic, and it is the reason the professor's slide-12 speaker note says simply "Quantity of real money balances = M/P ".

Link 2: $M/P \downarrow \Rightarrow i \uparrow$. This is the money-market equilibrium condition from the IS-LM chapter. Households and firms need money to carry out transactions; at a higher price level, the same volume of real transactions requires *more* money. At the original interest rate there is now an **excess demand for money**. To obtain money, people sell interest-bearing assets (bonds); a wave of bond selling pushes bond prices down, and a lower bond price *is* a higher interest rate. Equivalently: the interest rate is the opportunity cost of holding money, so it must rise until people are content to hold the smaller stock of real balances. In IS-LM language, the **LM curve shifts to the left / upward**.

Link 3: $i \uparrow \Rightarrow I \downarrow$. This is the investment function from the goods market. A higher interest rate raises the cost of borrowing to finance a project, and raises the return on the alternative of simply holding financial assets. Marginal projects that were just worth doing at the old rate are no longer worth doing. Investment spending falls. (Interest-sensitive consumption — cars, durables bought on credit, housing — falls for the same reason.)

Link 4: $I \downarrow \Rightarrow Y \text{ demanded} \downarrow$. Since $Y = C + I + G + NX$, lower investment means lower aggregate demand — and through the multiplier, the eventual fall in demand is larger than the initial fall in I .

MECHANISM — THE INTEREST-RATE EFFECT (LEARN THIS CHAIN VERBATIM)



Read in reverse for a fall in the price level. The inverse relation between P and Y that this chain produces **is** the downward-sloping AD curve.

Figure Deriving the Aggregate demand Curve

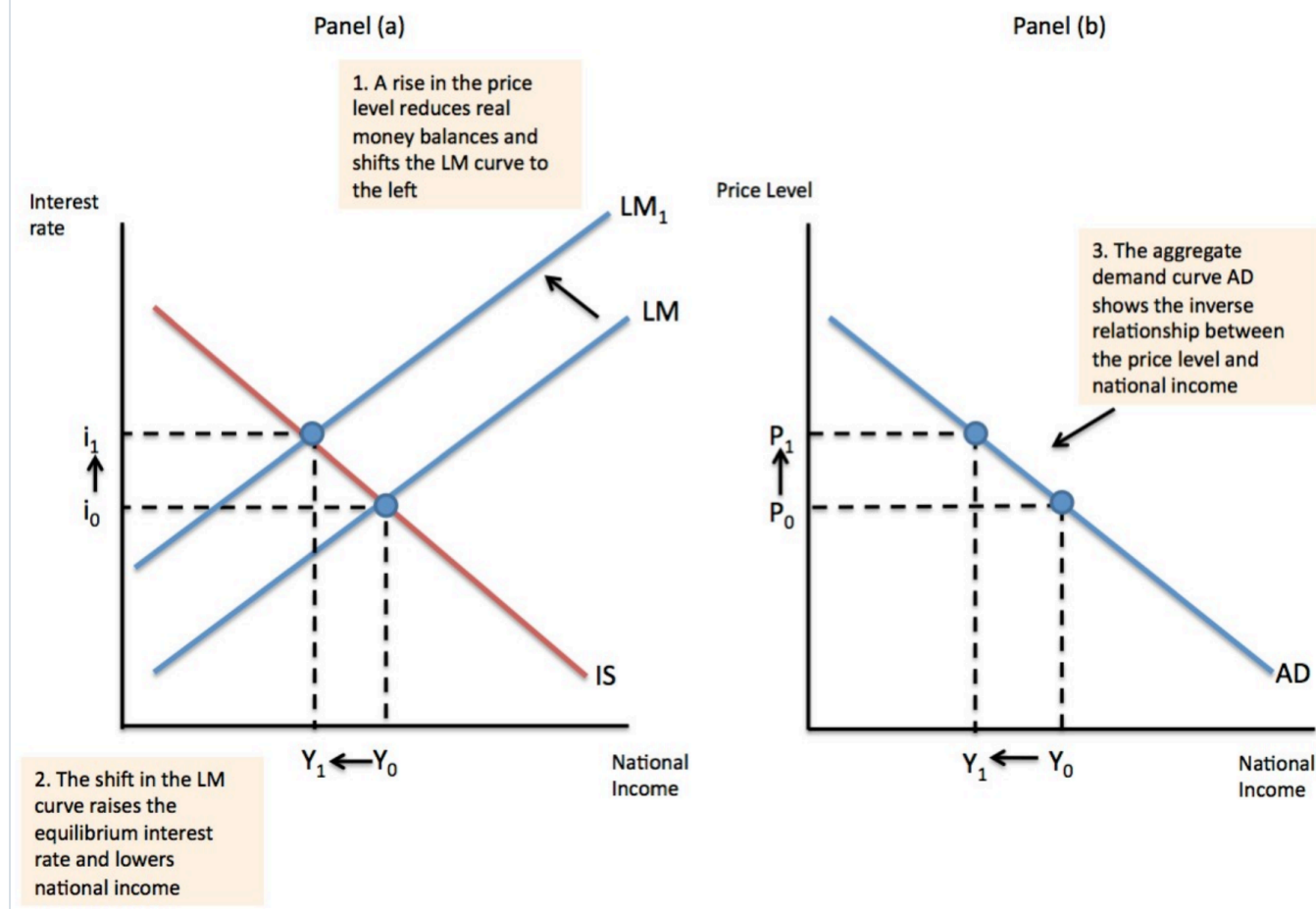


Figure (slide 12) — Deriving the aggregate demand curve from IS-LM. Panel (a) is the familiar IS-LM diagram, with the interest rate on the vertical axis and national income on the horizontal axis. Start at the intersection of IS and LM: interest rate i_0 , income Y_0 . Now let the price level rise from P_0 to P_1 . Because M is unchanged, real money balances M/P fall, and the LM curve shifts **left/upward** to LM_1 (callout 1). The economy slides up along the unchanged IS curve to the new intersection: the interest rate rises to i_1 and income falls to Y_1 (callout 2) — investment has been crowded out by the higher interest rate. **Panel (b)** simply re-plots the two results in a new diagram with the **price level** on the vertical axis: the pair (Y_0, P_0) and the pair (Y_1, P_1) . Joining them traces out the aggregate demand curve, which therefore shows the inverse relationship between the price level and national income (callout 3).

Exam tip — what this derivation buys you

Three examinable consequences follow immediately from "AD is the locus of IS-LM equilibria":

1. **Every point on the AD curve is a point at which both the goods market and the money market clear.** AD is not a "wish list" — it is a set of full IS-LM equilibria, one for each price level.
2. **Anything that shifts IS or LM, other than a change in P , shifts the whole AD curve.** A change in G , in taxes, in confidence, in the money supply M — all of these move AD. A change in P alone moves you *along* AD.

3. The interest rate is not on the AD-AS diagram, but it is moving in the background.

Every time the economy moves up along AD (higher P , lower Y), the interest rate is rising and investment is falling. Questions that ask what happens to i or to I are testing exactly this. Keep the hidden IS-LM diagram in your head.

Going deeper — what makes the AD curve flat or steep?

Not required by the slide, but it explains several MCQ answers. A rise in P shifts LM left by a given horizontal distance. How much output falls as a result depends on the rest of the model:

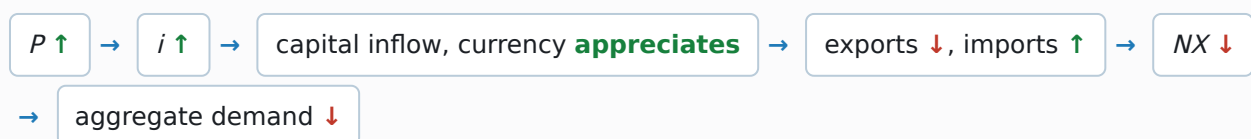
- **How interest-sensitive investment is** (the slope of IS). If investment barely responds to the interest rate, IS is steep, output hardly moves, and **AD is steep**. In the extreme where investment is completely interest-insensitive, the interest-rate effect vanishes and AD would be vertical (only the small wealth and exchange-rate effects would remain).
- **How interest-sensitive money demand is** (the slope of LM). If money demand is very interest-sensitive, a small rise in i restores money-market equilibrium, so a given fall in M/P barely raises i , and output barely falls: **AD is steep**. In a liquidity trap the interest rate cannot rise at all, and the interest-rate effect is switched off.
- **The size of the multiplier**. A larger multiplier turns a given fall in investment into a larger fall in output: **AD is flatter**.

Effect 3 — The exchange-rate effect (net exports) [bracketed on the slide]

The third channel works through the open-economy component NX , and it is really a continuation of the interest-rate effect. When the domestic price level rises, the interest rate rises (links 1-2 above). A higher domestic interest rate attracts financial capital from abroad: foreign investors buy domestic assets, and to do so they must buy the domestic currency. Demand for the currency rises, so it **appreciates**. An appreciated currency makes domestic goods more expensive for foreigners and foreign goods cheaper for domestic buyers, so exports fall, imports rise, and **net exports fall** — reducing aggregate demand.

There is also a direct route, without going through the interest rate: if domestic prices rise while foreign prices are unchanged, domestic goods become less competitive, so NX falls.

MECHANISM — THE EXCHANGE-RATE EFFECT



Exam tip — how to use the three effects in a written answer

A complete answer to "Why does the AD curve slope downward?" names all three effects, attaches each to **the component of GDP it works through**, and gives the mechanism for at least the interest-rate one:

- Wealth effect → **C** — lower P raises the real value of nominal assets, households feel richer, consumption rises.

- Interest-rate effect → **I** — lower P raises M/P , the interest rate falls, investment rises. (*Say explicitly that this is the IS-LM channel and that LM shifts.*)
- Exchange-rate effect → **NX** — the lower interest rate causes a depreciation, net exports rise.

Then finish with the sentence that ties it together: "so a lower price level is associated with a larger quantity of output demanded, which is why AD slopes downward." And add the discriminating remark: *G does not respond to the price level.*

§5 Why the AD curve shifts

SLIDE 13 — ON THE SLIDE

Why the Aggregate Demand Curve Might Shift

- The downward slope of the aggregate demand curve shows that a fall in the price level raises the overall quantity of goods and services demanded.
- Many other factors, however, affect the quantity of goods and services demanded at any given price level.
 - When one of these other factors changes, the aggregate demand curve shifts.

This slide exists purely to draw the line between **movement along** and **shift of**. It is worth being pedantic about, because it is the single most frequently penalised error in this chapter.

	Movement <i>along</i> AD	Shift <i>of</i> AD
Caused by	A change in the price level P — and nothing else	A change in any other determinant of C , I , G or NX at a given P
What is held constant	Confidence, taxes, G , M , foreign income — everything except P	P is held constant; the curve moves horizontally
In IS-LM terms	LM shifts because M/P changed via P ; you slide along the same AD	IS or LM shifts for a reason other than P ; the whole AD locus relocates
Language to use	"a change in the <i>quantity demanded</i> "	"a change in <i>aggregate demand</i> "

SLIDE 14 — ON THE SLIDE

Why the Aggregate Demand Curve Might Shift

Shifts arising from changes in

- **Consumption** — e.g. change in consumer confidence, change in income taxes and transfer payments, ...
- **Investment** — e.g. change in business confidence, change in monetary policy (interest rate effect), ...
- **Government Purchases** — e.g. change in fiscal policy, ...
- **Net Exports** — e.g. change in foreign income, ...

The organising principle is again $Y = C + I + G + NX$: to shift AD, something must change spending on one of the four components **at an unchanged price level**. Here is the expanded checklist, with the direction of the shift.

Determinants of aggregate demand — the full working list

Through	Change	Effect on AD and why
C	Consumer confidence rises (optimism about future jobs and incomes)	AD right. Households save less and spend more today out of the same current income.
	Income taxes cut, or transfer payments increased	AD right. Disposable income rises, consumption rises. (Fiscal policy acting through <i>C</i> , not through <i>G</i> .)
	Stock market boom / house prices rise	AD right. A wealth effect — but note this one is <i>not</i> caused by the price level, so it is a shift, not a movement along.
	Households decide to save more for retirement	AD left. Consumption falls at every price level.
I	Business confidence rises ("animal spirits", optimism about future profits)	AD right. Firms bring forward investment projects. Recall from §1 that this is the most volatile component — confidence swings move AD a lot.
	Expansionary monetary policy: the central bank increases <i>M</i> (or cuts the policy rate)	AD right. At an unchanged price level, a higher <i>M</i> means higher <i>M/P</i> , LM shifts right, <i>i</i> falls, <i>I</i> rises. This is the item the slide labels "monetary policy (interest rate effect)".
	Investment tax credit; corporate tax cut; new profitable technology	AD right. The return to investing rises, so more projects clear the hurdle rate.
G	Government increases spending on goods and services (infrastructure, defence, public wages); or cuts it under austerity	AD right (or left for a cut). Direct, first-round effect on demand, amplified by the multiplier. <i>G</i> is the purest shift variable in the model — it never responds to <i>P</i> .
NX	Foreign income rises (a boom in trading partners)	AD right. Foreigners buy more of our exports at every domestic price level.
	Currency depreciates for a reason unrelated to domestic <i>P</i> (e.g. capital outflows, a foreign interest-rate rise)	AD right. Exports become cheaper abroad, imports dearer at home.
	Trade barriers abroad; a recession in a major export market	AD left.

Trap — the "interest rate effect" appears twice and means two different things

Slide 11 uses "interest-rate effect" to explain the **slope** of AD: *P* changes, *M/P* changes, *i* changes → **movement along AD**. Slide 14 uses "monetary policy (interest rate effect)" to explain a **shift** of AD: *M* changes while *P* is unchanged, so *M/P* changes → **the whole AD curve moves**. Same economic channel (the money market), completely different diagrammatic consequence. The discriminating question is always: *what caused M/P to change — the numerator or the denominator?* Numerator (*M*, i.e. the central bank) ⇒ shift. Denominator (*P*) ⇒ movement along.

Shifts in the Aggregate Demand Curve

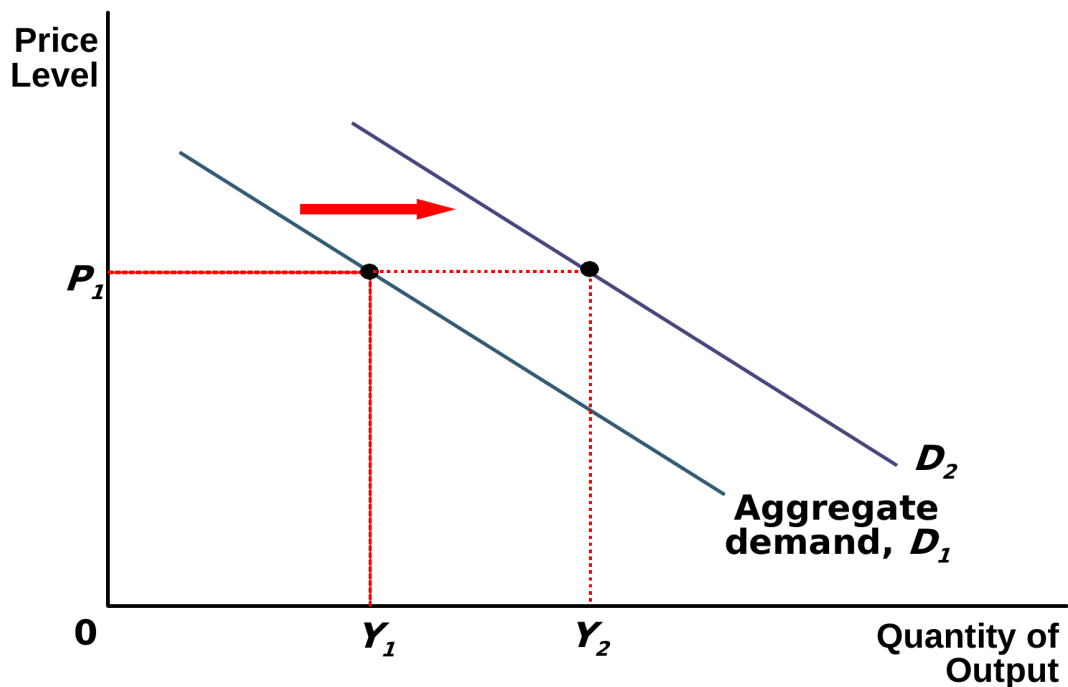


Figure (slide 15) — Shifts in the aggregate demand curve. The curve moves from D_1 to D_2 . Read the shift **horizontally**: at the unchanged price level P_1 , the quantity of output demanded rises from Y_1 to Y_2 . That is the correct way to describe a shift — "at every price level, more output is demanded." A rightward shift is called an **increase in aggregate demand**; a leftward shift is a **decrease in aggregate demand**. Note that this figure shows the shift *in isolation*: it deliberately says nothing about what happens to the equilibrium, because that also requires the AS curve. The full experiment is Figure 7 in §11.

Exam tip — describing a shift precisely

Say "at each price level, the quantity of goods and services demanded is higher" rather than "demand goes up". And be careful with the words *right* and *up*: for a downward-sloping curve, a rightward shift is also an upward shift, and either description is fine — but for the SRAS curve later, "left" and "up" also coincide, while "left" and "down" would be wrong. Get in the habit of describing shifts **horizontally, at a given price level**, and you will never mislabel one.

§6 Long-run aggregate supply

SLIDE 16 — ON THE SLIDE

The Aggregate Supply Curve

- In the long run, the aggregate supply curve is vertical.
- In the short run, the aggregate supply curve is upward sloping.
- The Long-Run Aggregate Supply Curve
 - In the long run, an economy's production of goods and services depends on its supplies of labour, capital, and natural resources and on the available technology used to turn these factors of production into goods and services.
 - The price level does not affect these variables in the long run.
 - Monetary neutrality in the long run

Aggregate supply is the one place in this model where the time horizon changes the *shape* of the curve, not just its position. That is unusual and it is the reason the chapter spends so long on it.

Why LRAS is vertical — the argument in full

Think about what an economy can physically produce over a long horizon. It has a stock of workers, a stock of machines and buildings, a set of natural resources, and a body of technological knowledge that determines how efficiently those inputs are combined. Write that as a production function:

$$Y = A \cdot F(K, L, N)$$

where K is capital, L labour, N natural resources and A the level of technology. Now ask the crucial question: **does the price level appear anywhere in that expression?** It does not. The number of workers is not a function of whether the price index is 100 or 200. Neither is the number of machines, nor the state of engineering knowledge.

So if you double every price and every wage overnight, the economy can still produce exactly the same basket of real goods and services. That is precisely **monetary neutrality** restated in supply-side language, and it is why the long-run aggregate supply curve is a **vertical line**: the quantity supplied is the same at any price level.

Going deeper — the labour-market version of the same argument

Here is why a vertical LRAS is consistent with the wage-bargaining story you will meet in §7. In the long run, expectations are correct: $P^e = P$. Workers bargain over the **real** wage W/P , and firms set prices as a mark-up over unit labour cost, which also pins down a real wage. When expectations are correct, these two real wages must be mutually consistent, and there is only one unemployment rate at which they are — the **natural rate of unemployment**. One unemployment rate means one level of employment, and one level of employment means one level of output. Notice that nothing in that argument mentions P itself: only *real* wages appear.

Hence output in the long run is independent of the price level. This is the bridge between \$6 and \$7, and it is worth one sentence in an open answer.

Figure 3 The Long-Run Aggregate Supply Curve

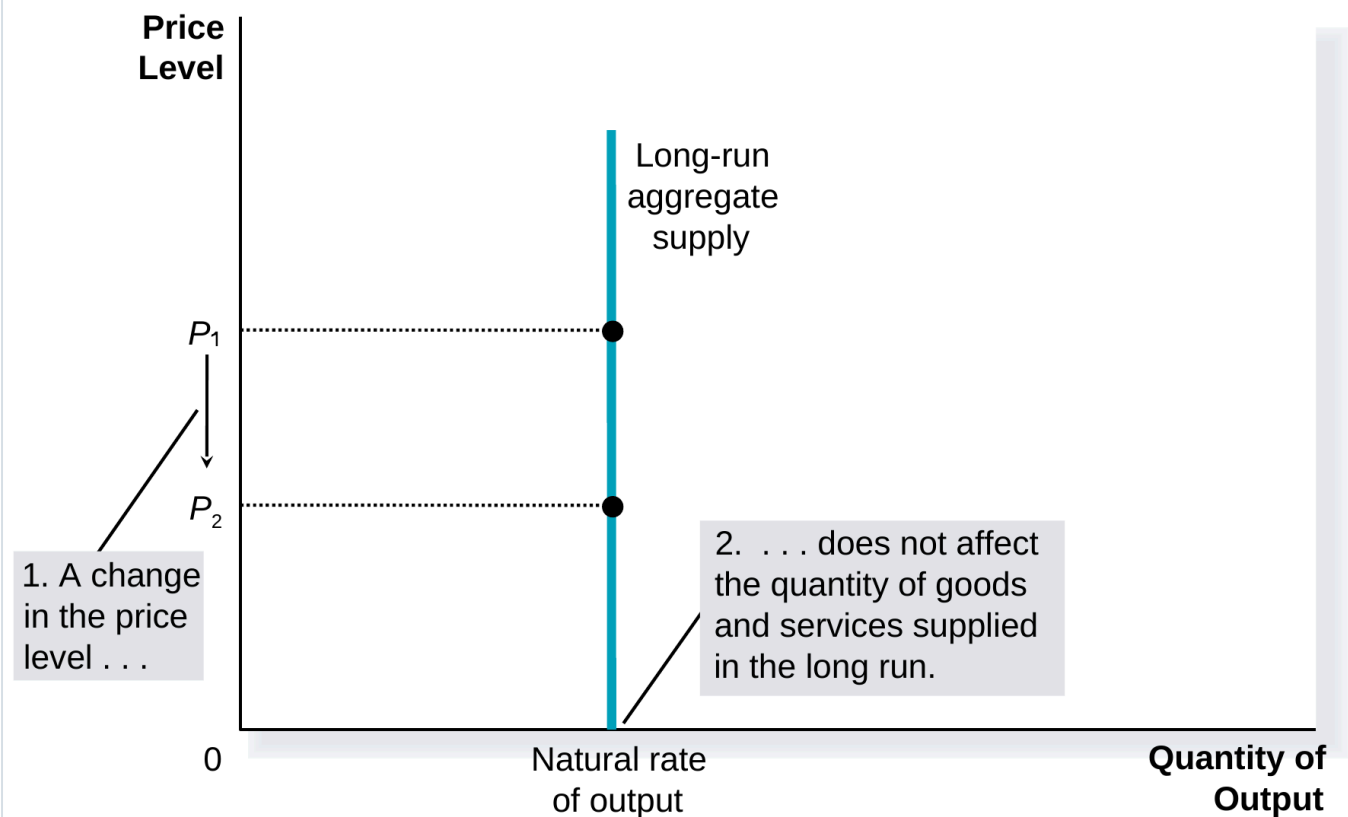


Figure 3 (slide 17) — The long-run aggregate supply curve. The curve is **vertical at the natural rate of output**. Callout 1: a change in the price level, here a fall from P_1 to P_2 ... Callout 2: ...does not affect the quantity of goods and services supplied in the long run. The economy simply slides *down the vertical line*: prices are lower, real output is unchanged. Note where the vertical line sits on the horizontal axis — at the **natural rate of output**, which is determined by labour, capital, natural resources and technology, and by nothing on the demand side.

SLIDE 18 — ON THE SLIDE

The Aggregate Supply Curve — the long-run curve

- The long-run aggregate supply curve is vertical at the natural rate of output.
- This level of production is also referred to as **potential output** or **full-employment output**.
 - The term "full-employment output" is a bit misleading: it is the level of production for which the unemployment rate is equal to the natural rate of unemployment.

Definition — learn this wording exactly

Natural rate of output (Y_n) — the level of production an economy achieves in the long run, when unemployment is at its natural rate. Also called *potential output* or *full-employment output*.

Natural rate of unemployment (u_n) — the unemployment rate towards which the economy gravitates in the long run; it consists of frictional unemployment (people between jobs, search and matching takes time) and structural unemployment (skill or geographical mismatch, and unemployment caused by wage-setting institutions such as minimum wages, unions and efficiency wages).

The professor explicitly flags that **"full-employment output" is misleading**, and that warning is exam-relevant. "Full employment" does *not* mean zero unemployment. Even at potential output there are always people between jobs and people whose skills do not match available vacancies. The natural rate is *natural* in the sense of "the rate the economy naturally returns to", not in the sense of "desirable" or "unavoidable" — it can and does change, for instance when labour-market institutions change (see §9).

Two further clarifications that come up in questions:

- **Output can exceed Y_n temporarily.** The natural rate is not a physical ceiling. In a boom, firms run overtime shifts, use capital more intensively, and unemployment falls below u_n . That situation is not sustainable — it puts upward pressure on wages and prices — but it is perfectly possible in the short run, and it is the case you get from a positive demand shock.
- **Y_n is not constant over time.** It grows as the labour force grows, as capital accumulates and as technology improves. That growth is the "long-run trend" that slide 4 says fluctuations happen *around*, and it is what Figure 4 draws.

SLIDE 19 — ON THE SLIDE

Why the Long-Run Aggregate Supply Curve Might Shift

- Any change in the economy that alters the natural rate of output shifts the long-run aggregate supply curve.
- The shifts may be categorized according to the various factors of production in the classical model that affect output.
 - Shifts arising from changes in: **Labour · Capital · Natural Resources · Technological Knowledge**

The four categories are just the four arguments of the production function. Note the logic: to move LRAS you must change the economy's **productive capacity**, or the **natural rate of unemployment**. Demand-side events never move LRAS.

Determinants of long-run aggregate supply

Factor	Examples that shift LRAS right	Examples that shift LRAS left
Labour	Population growth; net immigration; higher labour-force participation; a <i>fall</i> in the natural rate of unemployment (better job-matching services, lower structural mismatch, reform of unemployment benefits)	Emigration; ageing and falling participation; a <i>rise</i> in the natural rate of unemployment (e.g. institutions that weaken matching)
Capital	Accumulation of physical capital through investment; accumulation of human capital through education and training	Destruction of the capital stock (war, natural disaster); sustained under-investment
Natural resources	Discovery of new reserves; better access to imported resources	Permanent depletion of reserves; a permanent loss of access to an imported resource
Technological knowledge	Innovation; diffusion of better production methods; liberalisation of international trade, which acts like a technological improvement by letting the country specialise	Regulation or disruption that permanently lowers productivity

Trap — "a change in labour" is not "a change in employment"

A recession raises unemployment and lowers employment, but it does **not** shift LRAS. LRAS depends on the *supply* of labour and on the *natural* rate of unemployment, not on how many people happen to be employed this quarter. Cyclical unemployment is a movement away from LRAS, not a movement of LRAS. Only something that changes u_n itself — an institutional or structural change — moves the vertical line.

Putting growth and inflation together

Figure 4 Long-Run Growth and Inflation

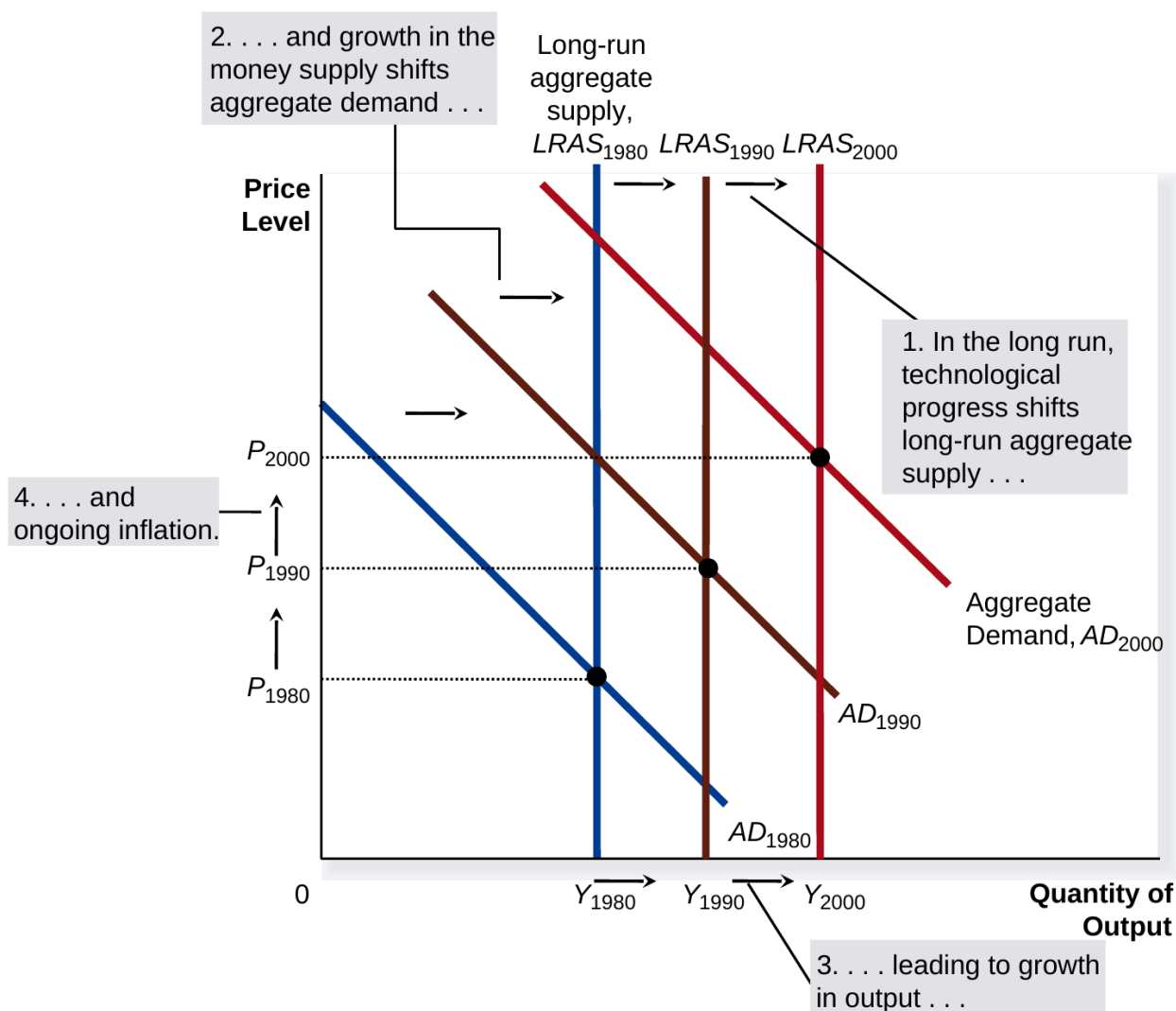


Figure 4 (slide 20) — Long-run growth and inflation. This figure shows the *trend* that the business cycle fluctuates around, drawn over decades rather than quarters. Callout 1: in the long run, technological progress (and capital accumulation, and labour-force growth) shifts long-run aggregate supply to the right, $LRAS_{1980} \rightarrow LRAS_{1990} \rightarrow LRAS_{2000}$. Callout 2: growth in the money supply shifts aggregate demand to the right, $AD_{1980} \rightarrow AD_{1990} \rightarrow AD_{2000}$. Callout 3: the result is growth in output, $Y_{1980} < Y_{1990} < Y_{2000}$. Callout 4: and ongoing inflation, $P_{1980} < P_{1990} < P_{2000}$. Note the clean division of labour: **the supply side delivers the growth in real output; the demand side (money growth) delivers the inflation.** That is exactly the classical long-run result.

Exam tip — what Figure 4 is really for

Two things, both examinable. **First**, it shows that this model is fully consistent with everything you learned about long-run growth and about the quantity theory of money: real growth comes from the supply side, sustained inflation comes from money growth. **Second**, it justifies a simplification used in the rest of the chapter. From slide 28 onwards, LRAS is drawn as a fixed vertical line and AD as a fixed curve, and we look only at *deviations* from the trend. In reality both curves are drifting right all the time; we suppress the drift so that the short-run story is

visible. If an exam question asks why a recession can occur even while the economy is growing, this is the answer: a "recession" is a fall *relative to trend*, and Figure 4 is the trend.

§7 Short-run aggregate supply: wage and price setting

SLIDE 21 — ON THE SLIDE

The Short Run AS relationship

- In the short run, there is a positive relationship between the level of prices and the quantity of goods and services supplied.
- The positive relationship between the level of prices and the quantity of goods and services supplied is explained by:
 - The Sticky Wage Theory
 - (The Sticky Price Theory)
 - (The Misperceptions Theory)

Exam tip — read the brackets again

The sticky wage theory is unbracketed and gets two full slides (22 and 23). It is the theory you must be able to explain in detail. The other two are bracketed: know the name, know the one-line intuition, be able to recognise them in an MCQ. Below, the sticky-wage theory is developed at length and the other two are summarised in a compact box.

All three theories share the same logical skeleton, and it is worth stating the skeleton before the details, because that is the part that is genuinely being tested: **in the short run something nominal is stuck or mis-perceived, so a change in the price level changes a real variable that firms care about, and firms respond by changing production.** In the long run nothing is stuck any more, the real variable returns to normal, and output returns to Y_n . That is why every one of the three theories implies both an upward-sloping SRAS and a vertical LRAS.

The wage- and price-setting framework

SLIDE 22 — ON THE SLIDE

Why the Aggregate Supply Curve Slopes Upward in the Short Run

- Wages and price setting
 - Nominal wages, as bargained by unions and employers, depend amongst others on the **expected price level** and the **unemployment rate**
 - Expected price level: workers are not interested in nominal wages (W), but in their real wage (W/P)
 - Real wage = the amount of goods and services that can be bought with a wage
 - Price level in the next period is unknown
 - Adaptive expectations based on inflation in the past
 - Unemployment rate: a higher unemployment rate decreases workers' bargaining power

This slide is the theoretical heart of the supply side. Unpack it in two halves.

Half 1 — Wage setting: what determines the bargained nominal wage?

Wages are not set by an auctioneer clearing a spot market. They are **bargained**, typically between unions and employers, typically for a period ahead (a year, two years, sometimes longer), and typically in **nominal** terms — the contract says "€3,200 per month", not "the equivalent of 40 baskets of goods per month".

Two variables drive the outcome:

(a) The expected price level, P^e . Workers do not care about the number on the payslip; they care about what it buys. That is the **real wage W/P** — the slide defines it for you as "the amount of goods and services that can be bought with a wage". But the bargain has to be struck *today* for a wage that will be paid *over the coming period*, and the price level over that coming period is not yet known. So negotiators must use a forecast, P^e . If they expect prices to be 3% higher, they must push for a nominal wage roughly 3% higher just to stand still. Hence: **higher expected price level $\Rightarrow$ higher bargained nominal wage.**

(b) The unemployment rate, u . Bargaining power depends on outside options. When unemployment is high, a worker who walks away faces a long search and a weak labour market, and the employer can easily replace them; when unemployment is low, vacancies are plentiful, workers can credibly threaten to leave, and employers must bid to retain staff. Hence: **higher unemployment $\Rightarrow$ weaker worker bargaining power $\Rightarrow$ lower bargained nominal wage.**

Going deeper — the wage-setting relation in symbols

Everything on slide 22 can be written compactly as

$$W = P^e \cdot F(u, z) \quad \text{with} \quad \partial F / \partial u < 0$$

The nominal wage is proportional to the *expected* price level (the "amongst others" of the slide is z : everything else that affects bargaining — union density, unemployment benefits, employment protection, minimum wages). Dividing by P^e shows the content of the relation: workers bargain over an **expected real wage**, and that expected real wage is lower when unemployment is higher. You are not required to produce this equation, but it makes every comparative-static question mechanical.

Half 2 — Price setting: how do firms turn wages into prices?

The slide's title says "*wages and price setting*", and the price-setting half is the step most students omit. Labour costs are the dominant cost of production for most firms. Firms set the price of what they sell as a **mark-up over unit costs**: cover the cost of producing one more unit, and add a margin that reflects the degree of competition in the market.

Going deeper — the price-setting relation in symbols

$$P = (1 + \mu) \cdot W / A$$

where μ is the mark-up (higher when competition is weaker) and A is labour productivity, so W/A is **unit labour cost**. The two relations combine into

$$P = (1 + \mu) \cdot P^e \cdot F(u, z) / A$$

Read this as the SRAS curve in disguise. It says the actual price level rises when: the expected price level rises ($P^e \uparrow$), unemployment falls (which raises F), the mark-up rises (weaker competition, or a cost shock such as oil raising the price of non-labour inputs), or productivity falls. Every SRAS shifter in §9 is one of these four terms — which is why that list is not arbitrary.

The sticky wage theory

SLIDE 23 — ON THE SLIDE

Why the Aggregate Supply Curve Slopes Upward in the Short Run

- The Sticky Wage Theory
 - Nominal wages are slow to adjust, or are "sticky" in the short run
 - Because expected prices only adapt slowly to actual prices
 - Contracts and wages are only revised every now and then
 - An increase in output
 - Increases employment and reduces unemployment
 - Lower unemployment increases bargained nominal wages
 - Firms set higher prices to cover these higher nominal wage costs
 - These higher prices do not immediately result in higher bargained nominal wages because of sticky price expectations and contracts

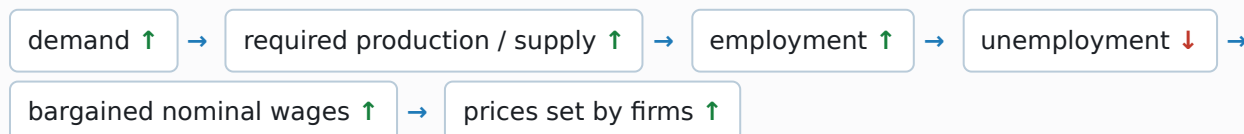
Two distinct things are being asserted here and you should keep them apart.

Assertion 1 — why wages are sticky. Two reasons, both on the slide:

- **Expectations adapt slowly.** Expectations are *adaptive* (slide 22: "based on inflation in the past"). If inflation has been 2% for years, negotiators will keep assuming roughly 2% even after something has changed. P^e only catches up gradually with P .
- **Contracts are infrequent.** Even if a union realised today that prices will be higher, the wage is fixed by an agreement that may run for another eighteen months. Renegotiation is costly and staggered across sectors, so the aggregate nominal wage level moves sluggishly.

Assertion 2 — how stickiness produces an upward-sloping SRAS. This is the causal chain in the second half of the slide, and it is repeated in the professor's own speaker note on the next slide. Note the direction it runs: it starts from a change in *demand/output* and ends at *prices*.

MECHANISM — THE PROFESSOR'S CHAIN (FROM THE SLIDE-24 LECTURE NOTE)



Higher output goes together with a higher price level $\Rightarrow$ the short-run aggregate supply curve slopes **upward**.

And here is the step that makes it a *short-run* relationship, i.e. the fourth sub-bullet: **those higher prices do not immediately feed back into higher bargained wages**, because expectations are backward-looking and contracts are not yet up for renewal. For a while, therefore, firms can produce more and charge more without their wage bill catching up. When the feedback does eventually arrive — which is §8 — the short-run curve starts to shift and the extra output is lost.

Going deeper — the textbook's version of the same theory, and why both are correct

Mankiw & Taylor tell the sticky-wage story running in the opposite direction, from prices to output: the nominal wage W is **locked by contract**; the price level P turns out higher than expected; therefore the **real wage W/P falls**; labour is now cheaper than firms had planned for; firms hire more and produce more. Higher $P =$ higher Y : an upward-sloping SRAS.

The professor's version runs from output to prices via bargaining; the textbook's runs from prices to output via the real wage. They are two readings of the same equilibrium relationship between P and Y , and both rely on the same friction — **nominal wages that do not immediately track the actual price level**. Being able to state both, and to say that they describe the same curve, is a strong exam answer. If you are asked to "explain the sticky wage theory", give the professor's chain first (it is what the slides say) and add the real-wage version as a second sentence.

Going deeper — the compact SRAS equation

The textbook summarises all three theories in one equation:

$$Y = Y_n + a \cdot (P - P^e), \quad a > 0$$

Output deviates from its natural rate only to the extent that the actual price level deviates from the expected price level. It is worth memorising because it makes four things instantly obvious:

- SRAS slopes **upward** in (Y, P) space, since $a > 0$.
- SRAS **crosses LRAS exactly where $P = P^e$** — expectations correct $\Rightarrow Y = Y_n$. This is why every SRAS curve you draw must intersect LRAS at the expected price level.
- A rise in P^e shifts SRAS **up/left** by exactly that amount (§8).
- The parameter a is the **flatness** of SRAS. It is large when nominal wages respond weakly to labour-market conditions, and small when they respond strongly. This is precisely what Statement 2 of slide 35 is about (§13).

The two bracketed theories — know the one-liners

Sticky price theory. Not all prices adjust instantly either, because of *menu costs*: reprinting catalogues, updating price lists, annoying customers with frequent changes. When aggregate demand falls, firms whose prices are temporarily stuck find themselves with prices above the new market-clearing level, so their sales fall, and they cut production. Hence a lower price level goes with lower output — an upward-sloping SRAS.

Misperceptions theory. Suppliers monitor the price of their *own* product much more closely than the general price level. When the overall price level falls unexpectedly, an individual wheat farmer sees the price of wheat fall and may wrongly conclude that the *relative* price of wheat has fallen — that wheat has become a bad business — and so produces less. Aggregate output falls with the price level even though nothing real has changed. Once producers realise all prices fell, they revert to normal production; hence the effect is short-run only.

Figure 5 The Short-Run Aggregate Supply Curve

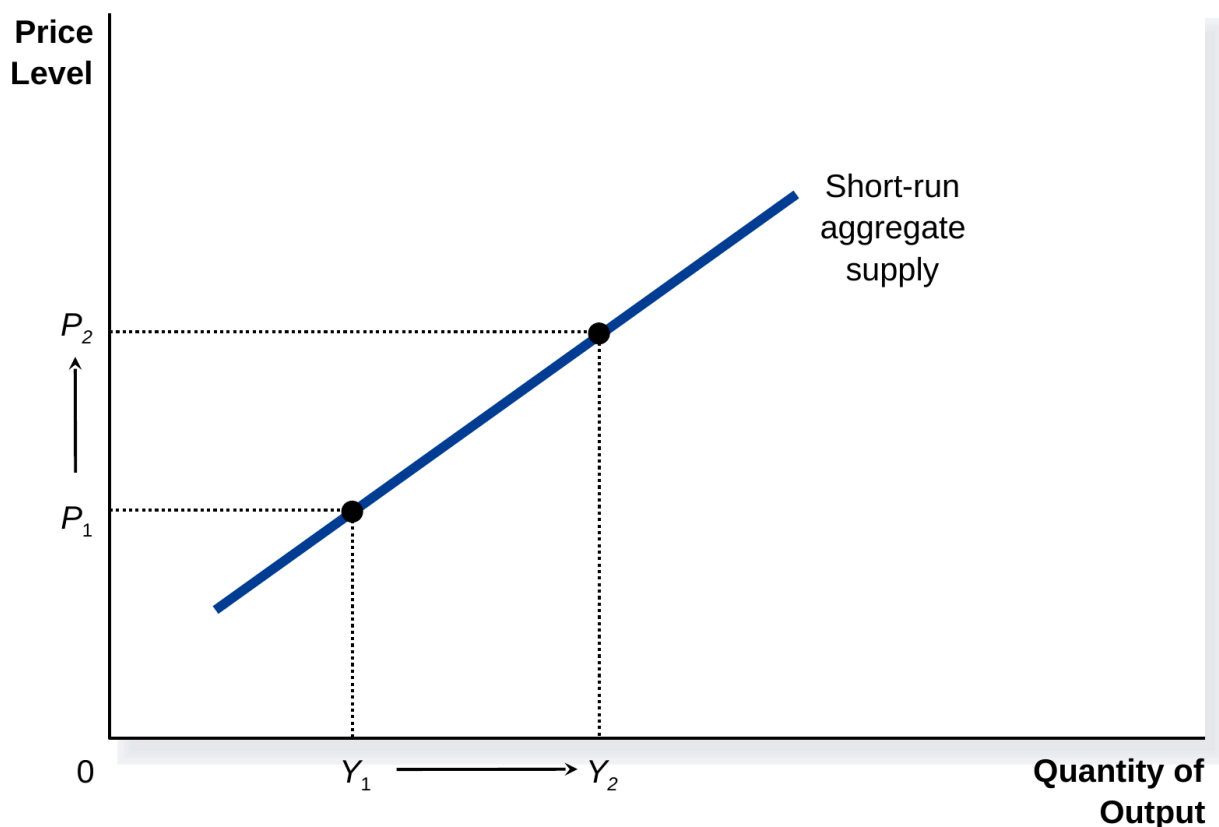


Figure 5 (slide 24) — The short-run aggregate supply curve. In the short run, a rise in the price level from P_1 to P_2 goes together with a rise in the quantity of output supplied from Y_1 to Y_2 . Two cautions. **(1)** This is a *movement along* SRAS, drawn for a **given expected price level** and given costs. **(2)** Do not read the arrow as pure causation from P to Y : in the professor's account the chain runs from higher demand $\rightarrow$ higher output $\rightarrow$ lower unemployment $\rightarrow$ higher bargained wages $\rightarrow$ higher prices set by firms. The curve records the *association* between the two; the mechanism is the wage-bargaining chain.

§8 From short-run to long-run equilibrium

SLIDE 25 — ON THE SLIDE

From short-run equilibrium to long-run equilibrium

- Over time, unions will adapt their expected prices to actual prices
 - An increase in expected prices increases bargained wages over time
 - For any given actual price level,
 - this reduces the quantity of goods and services that firms are willing to supply
 - and shifts the short-run aggregate supply curve to the left.
 - Or, different interpretation
 - firms set higher prices in response to higher production costs, which
 - leads to an upward shift of the short-run aggregate supply curve

SLIDE 26 — ON THE SLIDE

From short-run equilibrium to long-run equilibrium

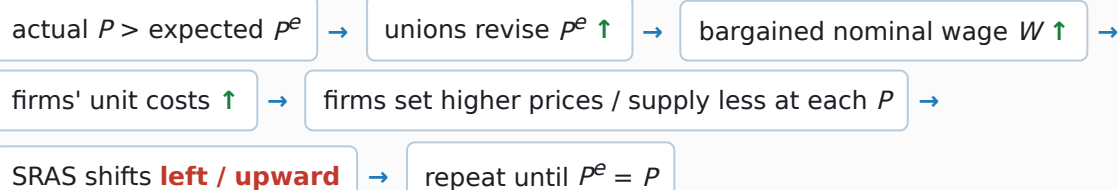
- Higher prices set by firms over time again translate in
 - Higher price expectations by unions, higher bargained wages and higher prices set by firms
 - Shift to the left / upward of the SRAS
- Long-run equilibrium is restored once (ex-ante) expected prices are equal to (ex-post) actual prices
 - No further need for unions to upwardly adapt expected prices
 - No further need for unions to upwardly adapt bargained nominal wages
 - No further need for firms to upwardly adapt the prices of their produced goods and services

These two slides describe the **adjustment process** — the engine that drives the economy from wherever a shock leaves it back to the natural rate. If you understand this section, Figures 7, 8 and 9 become trivial, because they are all the same process applied to different shocks.

The idea in one paragraph

The SRAS curve is drawn **for a given expected price level**. If the economy ends up at a point where actual prices are higher than wage setters expected, then wage setters have made a forecasting error: they accepted a nominal wage that turned out to deliver a lower real wage than they intended. They will not repeat the mistake. At the next round of bargaining they revise P^e upward, demand and obtain a higher nominal wage, firms' unit labour costs rise, and firms pass the higher costs into prices. Because the SRAS curve is drawn for the *old* P^e , this revision **shifts the whole SRAS curve left/upward**. And the process repeats: as long as actual prices keep exceeding expected prices, expectations keep being revised and SRAS keeps shifting. The process only stops when the forecast error is gone — when $P^e = P$. At that point output is at the natural rate, and the economy is on LRAS.

THE ADJUSTMENT LOOP (RUN IT IN REVERSE FOR A FALL IN PRICES)



The two interpretations on slide 25 are the same movement

The professor deliberately offers two readings, because students find one or the other more natural. They describe the **identical** shift of the curve:

Interpretation	How you read the shift	Sentence to write in an exam
Horizontal ("to the left")	Fix the price level on the vertical axis and ask how much firms are willing to supply. Higher wage costs make production less profitable at that price, so they supply less.	"At any given price level, higher nominal wages reduce the quantity of output firms are willing to supply, so SRAS shifts to the left."
Vertical ("upward")	Fix the quantity of output on the horizontal axis and ask what price firms need. Higher unit costs mean a higher mark-up price is needed to produce that same quantity.	"To produce any given level of output, firms must now charge a higher price to cover higher wage costs, so SRAS shifts upward."

For an upward-sloping curve, shifting left and shifting up are the same geometric operation. Use whichever wording is more convenient — but be consistent within one answer.

Definition — the long-run equilibrium condition

The economy is in **long-run equilibrium** when **(ex-ante) expected prices equal (ex-post) actual prices**: $P^e = P$. At that point there is no forecast error, so no reason for wage setters to revise expectations, no reason for nominal wages to change, and no reason for firms to change prices. Equivalently: $Y = Y_n$ and $u = u_n$, and all three curves (AD, SRAS, LRAS) pass through the same point.

Exam tip — the language "ex-ante" and "ex-post"

The slide uses these deliberately. **Ex-ante** = before the fact, at the moment the wage is bargained, when the price level is still a forecast. **Ex-post** = after the fact, once prices have actually been set and observed. The whole short run is the gap between the two; the long run is when the gap closes. If a question asks "what characterises long-run equilibrium in this model?", the highest-scoring single sentence is: *expected and actual prices coincide, so there is no further pressure on wages or prices, and output equals the natural rate.*

Trap — direction of the shift after a demand contraction

The slides state the adjustment for the case where actual prices *exceed* expectations, so SRAS shifts left. Make sure you can run it the other way. After a **fall** in aggregate demand, the

actual price level ends up *below* what wage setters expected. They then revise P^e **down**, accept **lower** nominal wages (because unemployment is high, their bargaining power is weak), firms' costs fall, and **SRAS shifts right / downward** until output is back at the natural rate. This is exactly what happens between points B and C in Figure 7. Getting this direction backwards is one of the most common errors in the exam.

§9 Why the SRAS curve shifts

SLIDE 27 — ON THE SLIDE

Why the Short-Run Aggregate Supply Curve Might Shift

- Apart from a change in **expected price level**, shifts may arise from changes in
 - **Labour** — e.g. change in the natural rate of unemployment
 - **Capital**
 - **Natural Resources** — e.g. changes in oil prices as a consequence of a decision of the OPEC cartel
 - **Technology**

The structure of this slide is: **the four LRAS shifters, plus one extra that belongs only to the short run**. That is the key to the whole section, so state it as a rule:

The rule to memorise

Anything that shifts LRAS also shifts SRAS in the same direction (a change in the economy's productive capacity changes both what it can produce eventually and what it produces now). **But a change in the expected price level shifts SRAS only** — it leaves LRAS untouched, because expectations are a nominal variable and LRAS depends only on real factors.

Determinants of short-run aggregate supply

Shifter	Shifts SRAS?	Shifts LRAS?	Comment
Expected price level P^e	Yes	No	Higher $P^e \rightarrow$ higher bargained wages $\rightarrow$ higher costs $\rightarrow$ SRAS left/up. This is the adjustment mechanism of §8. It is purely nominal, so productive capacity is unchanged.
Labour	Yes	Yes	The slide's own example is a <i>change in the natural rate of unemployment</i> . A higher u_n (say, from institutional change that weakens job matching) lowers the natural rate of output, moving both curves left.
Capital	Yes	Yes	More physical or human capital raises productivity, lowers unit costs, and raises potential output: both curves right.
Natural resources	Yes	Sometimes	The slide's example is an oil price rise engineered by OPEC . This raises firms' non-labour input costs, so at any output level firms need a higher price: SRAS shifts left/up. If the change is <i>temporary</i> , LRAS is unaffected; if it reflects a <i>permanent</i> loss of resource availability, potential output falls too and LRAS shifts as well.
Technology	Yes	Yes	Better technology raises productivity A , cuts unit labour cost W/A , and raises potential output: both curves right. A favourable supply shock.

Going deeper — mapping the list onto the price-setting equation

Recall $P = (1 + \mu) \cdot P^e \cdot F(u, z)/A$. Every entry in the table above is one term in that expression: P^e is the expectations shifter; the natural rate of unemployment and labour-market institutions sit in F and z ; technology and capital are A ; an oil-price shock raises non-labour costs, which enters as an increase in the effective mark-up μ (firms must charge more over labour cost to stay viable). This is why the list of shifters is exactly this list and not some other list — a nice sentence to include if an open question asks you to justify it.

Exam tip — the "which curve moves?" triage

Given any event, ask three questions in order:

1. **Does it change spending on C , I , G or NX at a given price level?** $\rightarrow$ AD shifts.
2. **Does it change firms' costs or wage setters' expectations, without changing productive capacity?** $\rightarrow$ SRAS only.
3. **Does it change the economy's productive capacity or the natural rate of unemployment?** $\rightarrow$ both SRAS and LRAS.

Worked examples: *a rise in oil prices* $\rightarrow$ 2 (SRAS left; LRAS unchanged if temporary). *A wave of immigration* $\rightarrow$ 3 (both right). *An increase in unemployment benefits that raises u_n* $\rightarrow$ 3 (both left). *The central bank raises the money supply* $\rightarrow$ 1 (AD right). *Unions revise inflation*

expectations upward → 2 (SRAS left only). *A stock-market crash* → 1 (AD left, via consumer wealth and confidence).

§10 The long-run equilibrium

Figure 6 The Long-Run Equilibrium

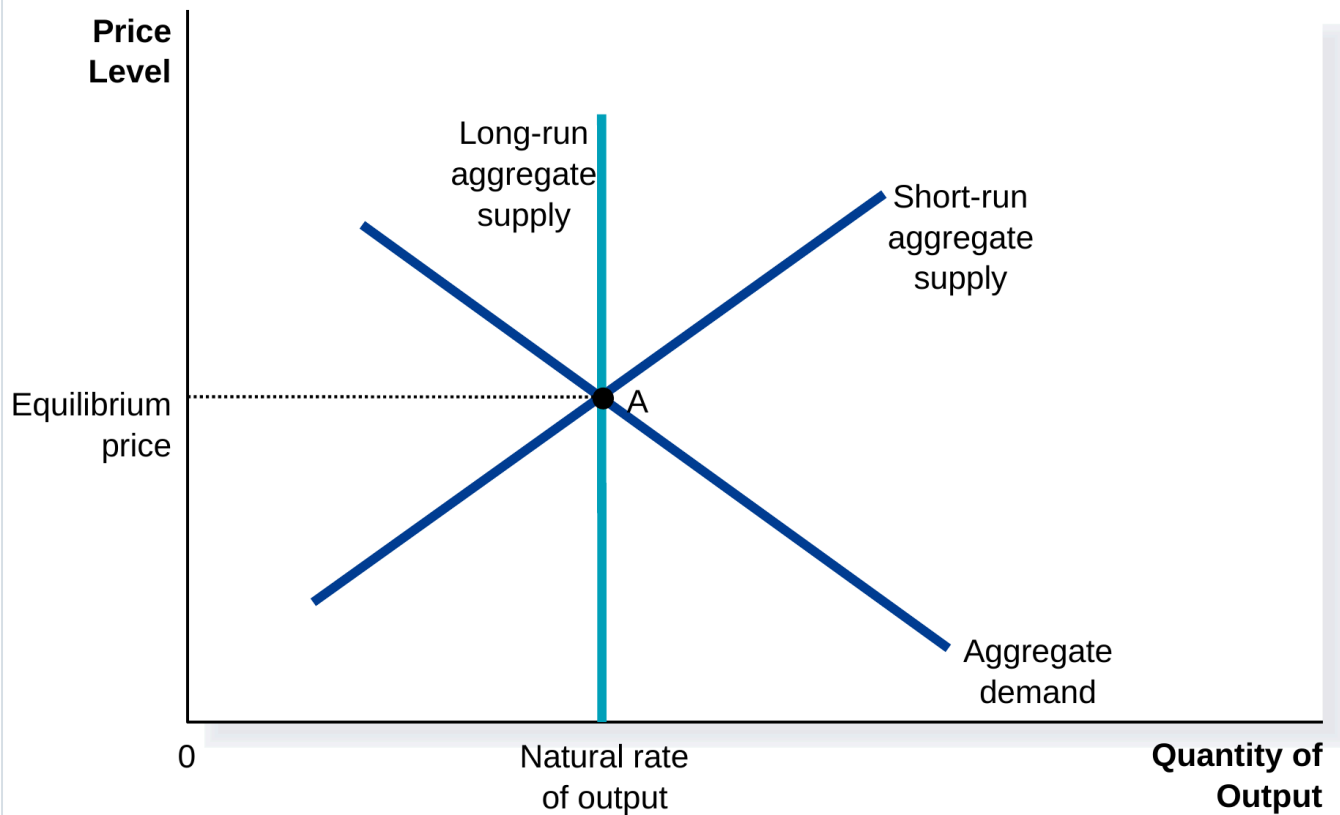


Figure 6 (slide 28) — The long-run equilibrium. All three curves pass through a single point **A**. Aggregate demand crosses short-run aggregate supply *exactly on* the vertical long-run aggregate supply curve, at the **natural rate of output** and the **equilibrium price**. This is the resting point of the model, and it is the starting point of every experiment that follows.

Point A is worth describing carefully, because "long-run equilibrium" is a bundle of four simultaneous conditions and exam answers are marked on how many of them you state:

#	Condition	What it means
1	$AD = SRAS$	The goods market clears at the current price level: what buyers want to buy equals what firms want to produce and sell. This is the <i>short-run</i> equilibrium condition, and it holds at every point in every diagram in this chapter.
2	$Y = Y_n$ (the intersection is on LRAS)	Output equals potential. The economy is producing what its factors of production and technology allow it to sustain.
3	$u = u_n$	Unemployment is at its natural rate. There is no cyclical unemployment — only frictional and structural.
4	$P^e = P$	Expectations are correct. Wage setters got their forecast right, so there is no pressure to revise wages, and hence no reason for SRAS to move. <i>This is the condition that makes the equilibrium a resting point rather than a passing moment.</i>

Exam tip — why condition 4 is the one that matters

Conditions 1–3 tell you *where* the economy is; condition 4 tells you *why it stays there*. At any point off the LRAS curve, at least one of the following is true: actual prices differ from expected prices, so expectations are being revised, so wages are changing, so SRAS is moving. The economy is therefore **not** at rest. Conversely, at A nothing is pushing anything, and the economy stays put until a new shock arrives. When an exam question asks you to "explain why the economy returns to the long-run equilibrium", the answer is condition 4 plus the adjustment loop from §8.

Trap — the SRAS curve you draw must go through the right point

When you draw a long-run equilibrium, the SRAS curve must cross LRAS **at the same point where AD crosses LRAS**. If you draw SRAS crossing LRAS somewhere else, you have drawn an economy where the goods market does not clear at the natural rate — an impossible long-run equilibrium — and you will lose marks even if your subsequent reasoning is fine. Practical drawing order: (1) draw the vertical LRAS; (2) draw AD crossing it; (3) draw SRAS through that same intersection point.

§11 Cause 1 — shifts in aggregate demand

SLIDE 29 — ON THE SLIDE

Two Causes of Economic Fluctuations

- Shifts in Aggregate Demand
 - In the short run, shifts in aggregate demand cause fluctuations in the economy's output of goods and services.
 - In the long run, shifts in aggregate demand affect the overall price level but do not affect output.

These two lines are the punchline of the whole chapter, so make sure you can see why each is true from the diagram. In the short run, SRAS is upward sloping, so when AD moves the intersection slides along a sloped curve and **both** P and Y change. In the long run, the economy ends up back on the **vertical** LRAS, so **only** P can differ. Monetary neutrality fails in the short run and reasserts itself in the long run — exactly as promised on slide 3.

Figure 7 A Contraction in Aggregate Demand

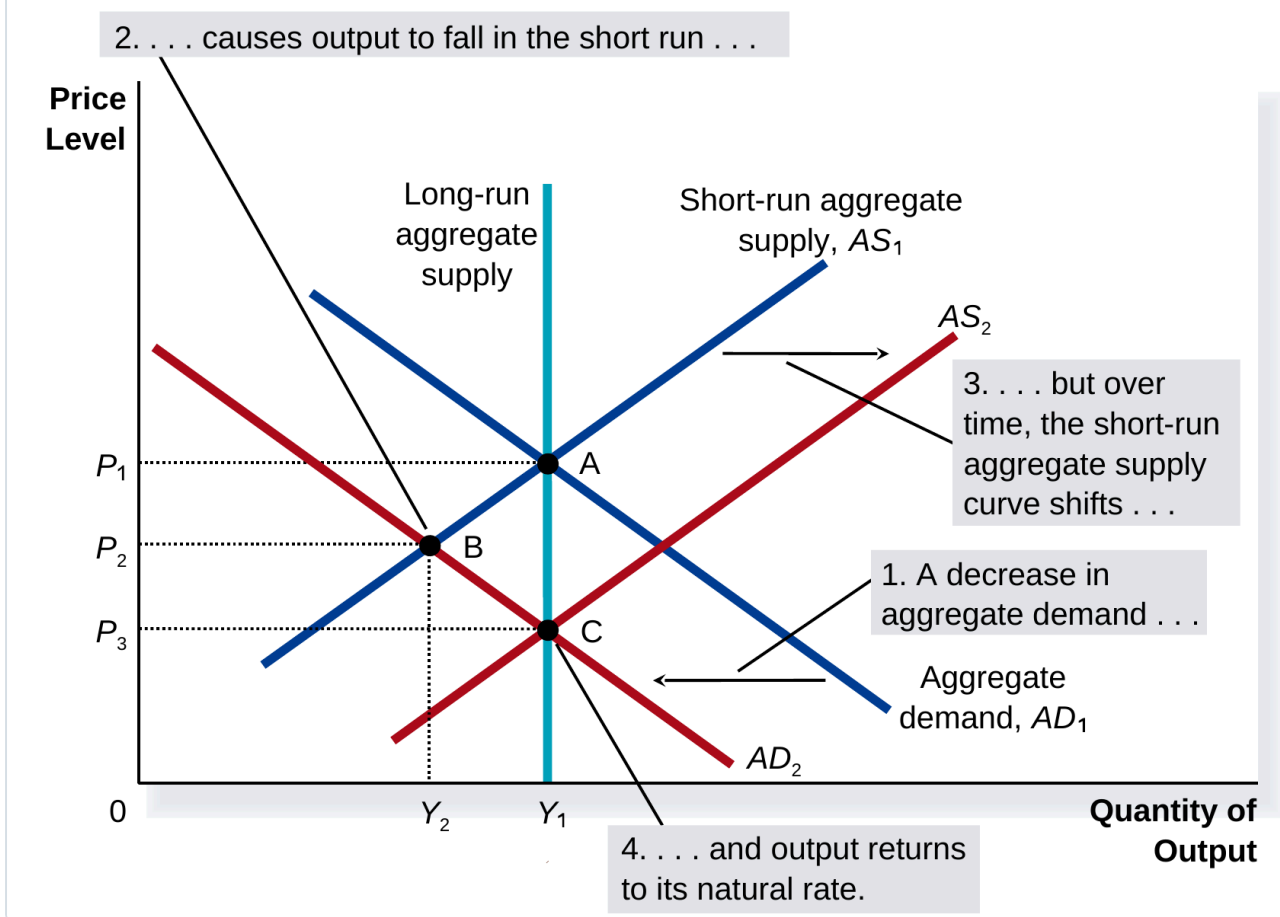


Figure 7 (slide 30) — A contraction in aggregate demand. Callout 1: a decrease in aggregate demand, $AD_1 \rightarrow AD_2$. Callout 2: causes output to fall in the short run — the economy moves from **A** down along the unchanged AS_1 to **B**, where output is $Y_2 < Y_1$ and the price level is $P_2 < P_1$. Callout 3: but over time the short-run aggregate supply curve shifts — $AS_1 \rightarrow AS_2$, to the right/downward, because expected prices and hence bargained wages fall. Callout 4: and output returns to its natural rate, at point **C**, with a still lower price level P_3 . Note that **A** and **C** are both on the LRAS line at Y_1 : the recession is temporary, the lower price level is permanent.

The experiment, step by step

Step 0 — the starting point. The economy is at **A**: a full long-run equilibrium. Output Y_1 is the natural rate, unemployment is at u_n , the price level is P_1 , and expectations are correct.

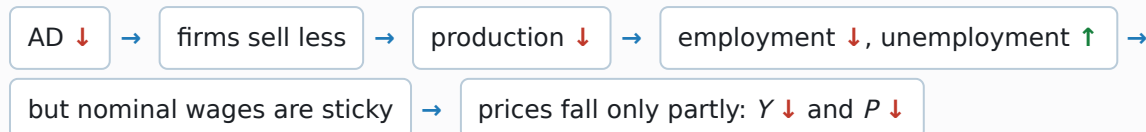
Step 1 — the shock. Something reduces desired spending at every price level: a collapse in consumer or business confidence, a stock-market crash that destroys household wealth, a fall in the money supply, fiscal austerity, or a recession abroad that cuts export demand. AD shifts left, $AD_1 \rightarrow AD_2$.

Step 2 — the short run (A → B). Wages are locked in by contracts and expectations have not yet been revised, so SRAS does not move. The economy slides *down along* AS_1 to **B**. Read off what happens:

- **Output falls** from Y_1 to Y_2 — a recession.

- **Unemployment rises** above the natural rate (Fact 3 of §1): with less output required, firms need fewer workers.
- **The price level falls** from P_1 to P_2 , but only partially — this is what an upward-sloping SRAS *means*. If SRAS were vertical, prices would absorb the whole shock and output would not move at all.

THE SHORT-RUN MECHANISM AT B (WRITE THIS OUT IN AN OPEN QUESTION)



Step 3 — the adjustment (B → C). At B the actual price level P_2 is **lower than wage setters had expected**, and unemployment is high. Both facts push in the same direction:

- expectations of prices are revised **downward**;
- high unemployment **weakens workers' bargaining power** (slide 22), so unions settle for lower nominal wage growth.

Lower bargained nominal wages reduce firms' unit costs, so at any given price level firms are willing to supply more: **SRAS shifts right/downward, $AS_1 \rightarrow AS_2$** . The economy slides *down along the new AD_2* from B to C.

Step 4 — the new long run at C. Output is back at the natural rate Y_1 ; unemployment is back at u_n ; the price level has fallen further, to P_3 ; and expectations are once again correct, so nothing moves any more. **The demand shock had a temporary effect on output and a permanent effect on the price level.** That is the classical result, recovered.

Summary of a contraction in aggregate demand

	Output Y	Price level P	Unemployment u	Which curve has moved
A → B (short run)	falls	falls	rises above u_n	AD only (left)
B → C (adjustment)	rises back	falls further	falls back to u_n	SRAS (right/down), as P^e and wages fall
A → C (net, long run)	unchanged	lower	unchanged	AD left, SRAS right; economy back on LRAS

Exam tip — the policy question that always follows

Figure 7 shows the economy healing itself, but slowly and painfully: the whole journey from B to C is a period of above-natural unemployment. That is why the alternative exists. Instead of waiting for wages and expectations to fall, **policymakers can shift AD back to the right** — an expansionary monetary policy (increase M , cut the policy rate) or an expansionary fiscal policy (raise G , cut taxes) — returning the economy directly from B to the original point A, with output at Y_1 and the price level back at P_1 . The trade-off to mention: self-correction is

slow and costly in lost output and unemployment, but active policy operates with lags and may overshoot. This is the "do nothing versus act" choice that reappears on slide 33.

Trap — do not shift SRAS at the moment of the shock

In the short run **only the shocked curve moves**. A demand shock moves AD and nothing else; SRAS moves only *later*, as a **consequence** of the forecast error the shock created. Students who shift both curves at once get an indeterminate answer and lose the entire mechanism. The correct sequencing is: shock → move one curve → read off the short run → *then* ask what the forecast error does to expectations → move SRAS → read off the long run.

Going deeper — what is happening to the interest rate along the way?

Remember that the AD curve hides an IS-LM diagram (§4). Between A and B the price level falls, so real money balances M/P **rise**, the LM curve shifts right and the interest rate **falls** — which cushions the recession by supporting investment. That is the movement *along* AD_2 . But the shock itself (say a collapse in confidence) shifted **IS left**, which lowers the interest rate too. So in a demand-driven recession, output and interest rates fall together. Contrast this with the supply shock in §12, where output falls but the interest rate *rises* — a difference that MCQs love.

§12 Cause 2 — shifts in aggregate supply, stagflation and policy

SLIDE 31 — ON THE SLIDE

Two Causes of Economic Fluctuations

- An Adverse Shift in Aggregate Supply
 - A decrease in one of the determinants of aggregate supply shifts the curve to the left:
 - Output falls below the natural rate of employment.
 - Unemployment rises.
 - The price level rises.

Note on the slide's wording

The first sub-bullet reads "output falls below the natural rate of *employment*". Read this as "**output falls below the natural rate of output**" — that is what is meant, and it is what the diagram shows. Use the correct term in your own answers: output falls below Y_n , and correspondingly unemployment rises above u_n .

Figure 8 An Adverse Shift in Aggregate Supply

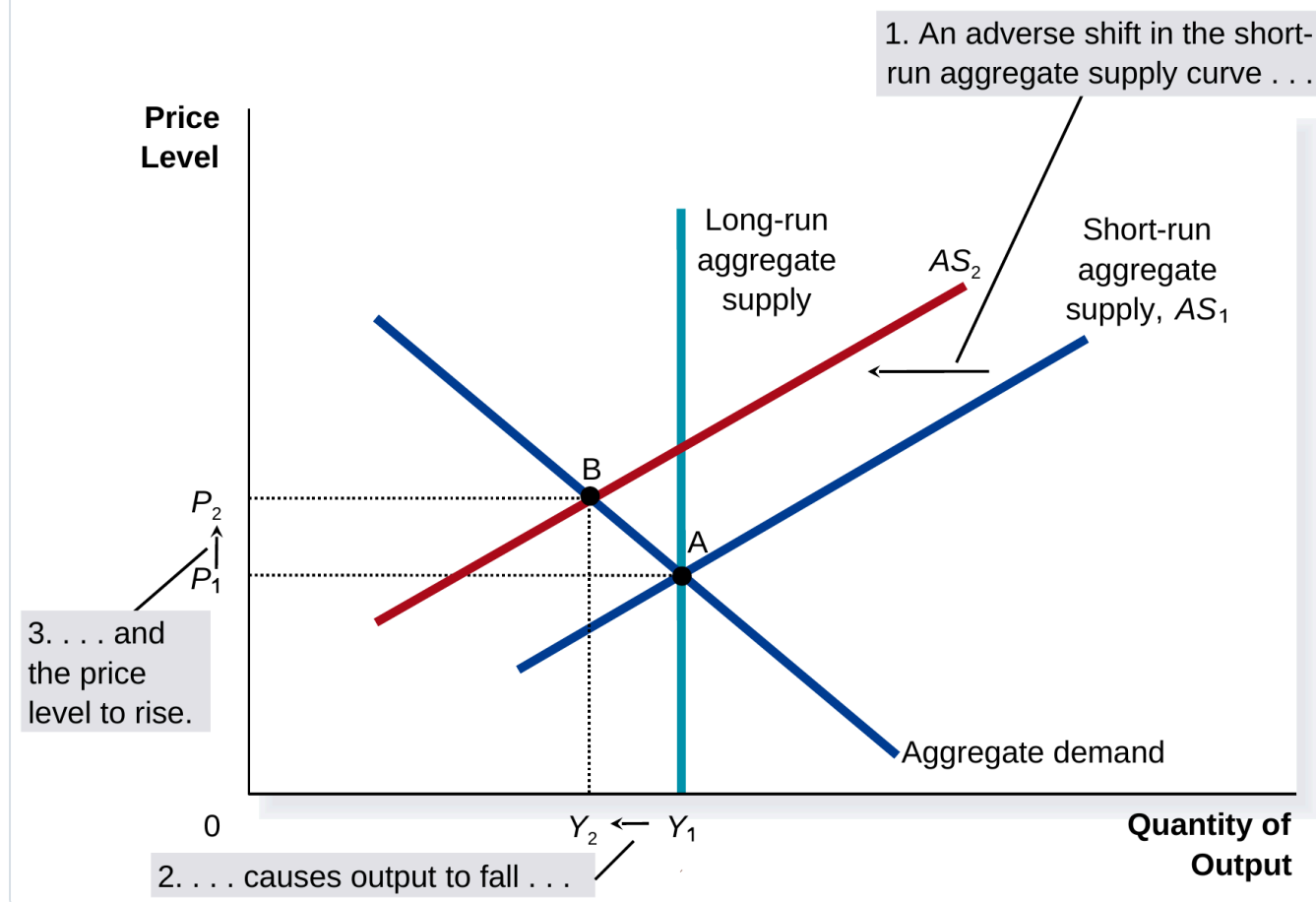


Figure 8 (slide 32) — An adverse shift in aggregate supply. Callout 1: an adverse shift in the short-run aggregate supply curve, $AS_1 \rightarrow AS_2$ (left / upward). The economy moves from **A** — a long-run equilibrium on the LRAS line — up along the unchanged aggregate demand curve to **B**. Callout 2: this causes output to fall, $Y_1 \rightarrow Y_2$. Callout 3: and the price level to rise, $P_1 \rightarrow P_2$. Output is now to the **left of LRAS**, so unemployment is above its natural rate — while prices are simultaneously higher. That combination is what makes a supply shock so much more awkward than a demand shock.

Why a supply shock produces the "wrong" combination

Take the classic example the professor gives on slide 27: **OPEC restricts supply and the oil price jumps**. Oil is an input into almost everything — transport, plastics, fertiliser, heating, electricity. Firms' costs per unit of output rise. Two equivalent ways to say what happens next:

- *Horizontally*: at any given selling price, production is less profitable, so firms supply less. SRAS shifts **left**.
- *Vertically*: to produce any given quantity, firms must now charge more to cover costs. SRAS shifts **up**.

Because AD has not moved, the economy travels **up along AD** from A to B. Moving up along a downward-sloping curve necessarily means output falls while the price level rises. Hence the two "bad" things happen at the same time — and that is the definition of stagflation.

SLIDE 33 — ON THE SLIDE

The Effects of a Shift in Aggregate Supply

- Stagflation
 - Adverse shifts in aggregate supply cause stagflation — a period of recession and inflation.
 - Output falls and prices rise.
 - Policymakers who can influence aggregate demand cannot offset both of these adverse effects simultaneously.
- Policymakers may respond to a recession in one of the following ways:
 - Do nothing and wait for price expectations and wages to adjust.
 - Take action to increase aggregate demand by using monetary and fiscal policy.

Definition

Stagflation — a period of falling output (stagnation, with rising unemployment) combined with a rising price level (inflation). It is the characteristic consequence of an adverse aggregate supply shock.

The policy dilemma, spelled out

The sentence "policymakers who can influence aggregate demand cannot offset both of these adverse effects simultaneously" is the examinable core of slide 33. The reason is **geometric** and worth stating that way: monetary and fiscal policy can only move **AD**, and AD is downward sloping, so moving it always pushes P and Y in the **same** direction. But the supply shock has moved them in **opposite** directions. One instrument cannot fix two problems that point opposite ways.

The three responses to an adverse supply shock

Response	What is done	Consequence
1. Do nothing (wait it out)	No policy change. Let the economy self-correct.	Output is below Y_n and unemployment above u_n . High unemployment weakens workers' bargaining power and, once the temporary cost shock passes, expectations of prices come back down; bargained wages fall; costs fall; SRAS shifts back right until the economy returns to the original point A, with the original output <i>and</i> the original price level. <i>Cost: a prolonged recession.</i>
2. Accommodate (expand AD)	Expansionary monetary or fiscal policy shifts AD right.	Output is brought back to the natural rate quickly, so the recession ends — but the price level ends up higher still . Inflation is validated rather than resisted. This is Figure 9.
3. Fight the inflation (contract AD)	Contractionary monetary or fiscal policy shifts AD left.	The price level is pushed back down towards P_1 , but output falls even further below the natural rate and unemployment rises further. Disinflation is bought at the price of a deeper recession. <i>(Not drawn on the slides, but the symmetric case — mention it to show you understand the dilemma.)</i>

Figure 9 Accommodating an Adverse Shift in Aggregate Supply

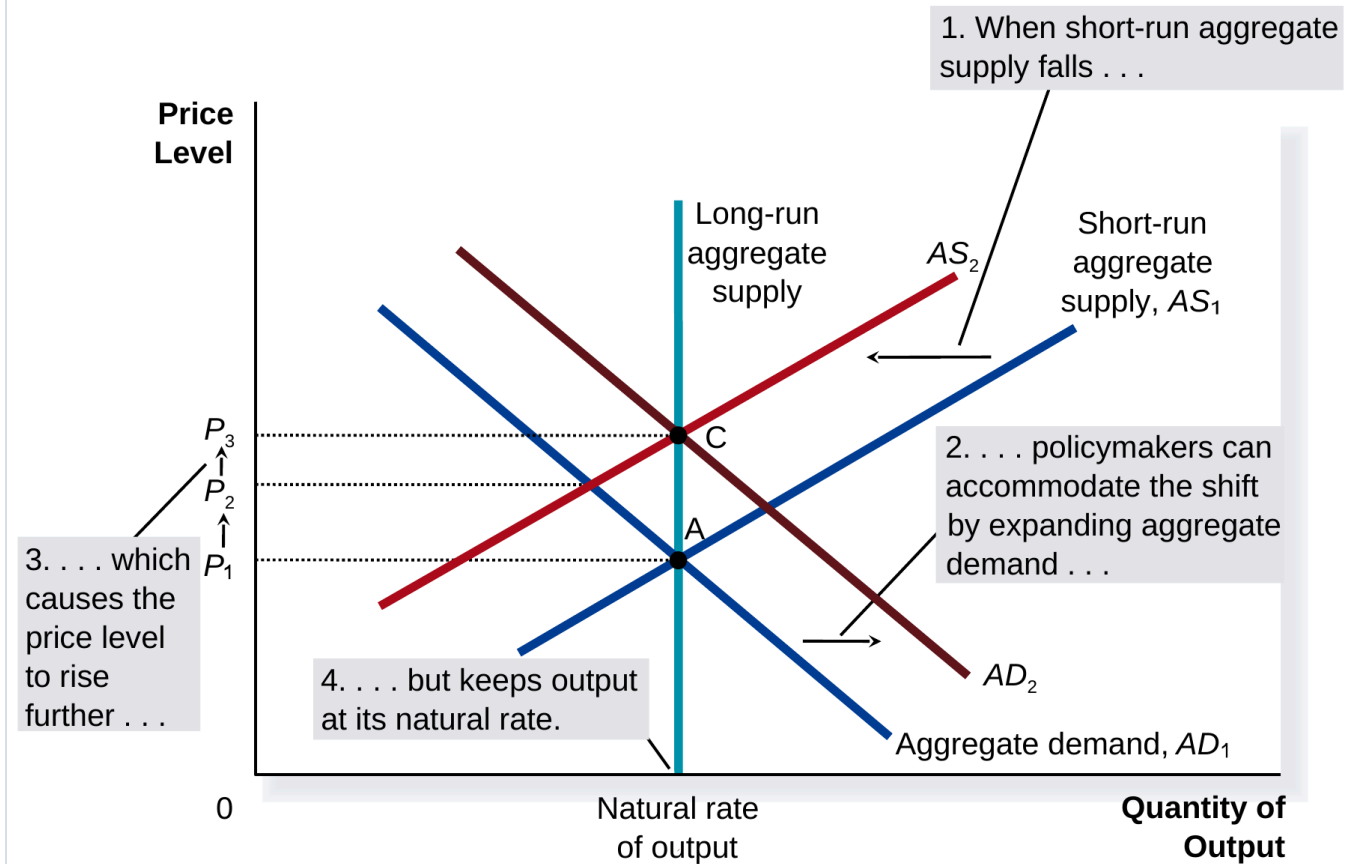


Figure 9 (slide 34) — Accommodating an adverse shift in aggregate supply. Callout 1: short-run aggregate supply falls, $AS_1 \rightarrow AS_2$, moving the economy from **A** towards a stagflationary point with output below the natural rate and a price level of P_2 . Callout 2: policymakers accommodate the shift by expanding aggregate demand, $AD_1 \rightarrow AD_2$. Callout 3: which causes the price level to rise further, to P_3 . Callout 4: but keeps output at its natural rate, at point **C**. Compare the vertical positions: $P_3 > P_2 > P_1$. The recession has been eliminated; the inflation has been made worse.

Definition

Accommodating a supply shock — responding to an adverse shift in aggregate supply by expanding aggregate demand, so that output is restored to its natural rate while the price level is permanently higher.

Exam tip — the danger of accommodation, in one sentence

A high-scoring extra remark: accommodation risks **entrenching higher inflation expectations**. If wage setters observe that every cost shock is validated by an expansion of demand, they will build higher expected inflation into future wage bargains, which shifts SRAS left again on its own — and the central bank must then either accommodate again (a wage-price spiral) or accept a recession later. This is the standard argument for why central banks care so much about "anchoring" inflation expectations, and it connects directly to slides 22 and 25–26: P^e is the variable that carries the shock forward in time.

Master comparison — demand shock versus supply shock

	Adverse demand shock (AD left)	Adverse supply shock (SRAS left)
Short-run output	Falls	Falls
Short-run price level	Falls	Rises
Short-run unemployment	Rises	Rises
Name for the episode	Recession / deflationary slump	Stagflation
Interest rate (hidden IS-LM)	Falls: P is lower so M/P is higher, plus IS has shifted left	Rises : P is higher so M/P is lower, LM shifts left
Investment	Falls (lower Y , though lower i cushions it)	Falls unambiguously (lower Y and higher i)
Self-correction	Expectations and wages fall; SRAS shifts right; output back at Y_n , price level permanently lower	Expectations and wages fall (once the cost shock passes); SRAS shifts back right; output back at Y_n , price level back at P_1
Can AD policy fix it?	Yes, fully — shifting AD back restores both output and the price level simultaneously	No — one instrument, two problems pointing in opposite directions; you must choose

§13 The two statements — fully worked

SLIDE 35 — ON THE SLIDE

Two statements...

Are the following statements true or false?

- **Statement 1:** Assume investment spending is determined by $I = f(i, Y)$. When an economy experiences stagflation, investment spending will fall in the short run.
- **Statement 2:** The higher the sensitivity of bargained nominal wages to unemployment, the lower the short-run impact of an increase in consumer confidence on GDP.

These are exactly the format of the exam's open true/false questions: a claim that requires you to identify the shock, work through the model, and check every channel. Below, each is worked in full.

Statement 1 — TRUE

Step 1 — decode the investment function. $I = f(i, Y)$ says investment depends on two things:

- **the interest rate i , negatively** ($\partial I / \partial i < 0$): a higher cost of borrowing kills marginal projects;
- **output/income Y , positively** ($\partial I / \partial Y > 0$): the accelerator effect — when sales and capacity utilisation are high, firms need more capital and expect higher profits, so they invest more.

So to sign the change in I , we need the signs of Δi and ΔY .

Step 2 — identify the shock. "Stagflation" means an **adverse shift of the SRAS curve** (Figure 8): the economy moves from A up along an unchanged AD curve to B, with output falling from Y_1 to Y_2 and the price level rising from P_1 to P_2 .

Step 3 — channel one: the output channel. Output has **fallen**. Since $\partial I / \partial Y > 0$, lower Y reduces investment. First channel: $I \downarrow$.

Step 4 — channel two: the interest-rate channel. This is the step that requires you to remember that the AD curve is an IS-LM construct. The economy has moved **up along the AD curve**, and moving up along AD is precisely the interest-rate effect of §4 running in reverse:

WHY THE INTEREST RATE RISES DURING STAGFLATION

$P \uparrow$ (to P_2) $\rightarrow$ real money balances $M/P \downarrow$ $\rightarrow$ LM shifts left $\rightarrow$ interest rate $i \uparrow$

Since $\partial I / \partial i < 0$, the higher interest rate also reduces investment. Second channel: $I \downarrow$.

Step 5 — conclusion. Both arguments of the investment function move in the direction that reduces investment: output is lower *and* the interest rate is higher. There is no offsetting

channel. Investment spending therefore falls unambiguously in the short run. **The statement is TRUE.**

MODEL ANSWER — WRITE IT LIKE THIS

True. Stagflation is caused by an adverse shift in short-run aggregate supply. The economy moves up along the unchanged AD curve, so output falls and the price level rises. With $I = f(i, Y)$, both arguments push investment down. (i) The *output channel*: lower Y reduces investment through the accelerator effect, since $\partial I / \partial Y > 0$. (ii) The *interest-rate channel*: the higher price level reduces real money balances M/P ; the LM curve shifts left; the equilibrium interest rate on the money market rises; and since $\partial I / \partial i < 0$, higher i reduces investment. Because both effects work in the same direction, investment spending falls unambiguously in the short run.

Exam tip — why the question specifies $I = f(i, Y)$

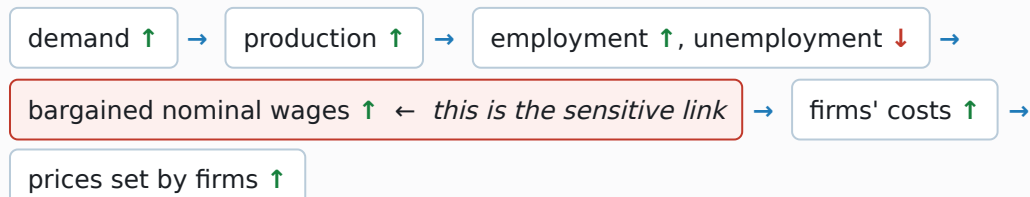
The specification is a deliberate hint that the examiner wants **both** channels, not one. If investment depended only on the interest rate, you would get full marks for the money-market argument alone. By writing Y into the function, the question is telling you: "there is a second channel, find it." Whenever a question hands you a functional form, treat each argument as a separate paragraph of your answer.

Statement 2 — TRUE

Step 1 — identify the shock. An increase in **consumer confidence** raises consumption at every price level, so **AD shifts right** (slide 14). In the short run the economy moves up along the SRAS curve: both P and Y rise. The question asks about the size of the increase in Y — the "short-run impact on GDP".

Step 2 — what "the sensitivity of bargained nominal wages to unemployment" controls. Go back to slide 22: nominal wages depend on the expected price level and on the unemployment rate, with higher unemployment lowering wages. The *sensitivity* in question is how strongly W responds to a given change in u . Now run the professor's mechanism chain from slide 23/24 with that sensitivity in mind:

THE SRAS MECHANISM, WITH THE SENSITIVITY HIGHLIGHTED



Step 3 — translate sensitivity into the slope of SRAS. The SRAS curve records how much the price level rises when output rises. If wages are **highly sensitive** to unemployment, then even a small increase in output — a small fall in unemployment — triggers a large increase in bargained wages, hence a large increase in costs, hence a large increase in prices. A small ΔY

produces a large ΔP : the SRAS curve is **steep**. Conversely, if wages barely react to labour-market conditions, output can expand a long way before prices move much: SRAS is **flat**.

The same point in the compact notation

Using $Y = Y_n + a(P - P^e)$: high wage sensitivity to unemployment means a **small a** (steep SRAS), low sensitivity means a **large a** (flat SRAS). In the limiting case of infinite sensitivity, $a \rightarrow 0$ and SRAS becomes vertical — the classical case, where demand shocks have no effect on output at all, even in the short run.

Step 4 — combine with the AD shift. A given rightward shift of AD is a fixed horizontal distance. How that distance is divided between more output and higher prices depends entirely on the slope of the curve it runs into:

- **Steep SRAS** (high wage sensitivity): the new intersection is reached after only a small increase in output and a large increase in the price level. **Small effect on GDP.**
- **Flat SRAS** (low wage sensitivity): the new intersection requires a large increase in output and only a small increase in the price level. **Large effect on GDP.**

Step 5 — conclusion. Higher sensitivity of bargained nominal wages to unemployment $\Rightarrow$ steeper SRAS $\Rightarrow$ a given rightward AD shift raises output by less (and prices by more). The short-run impact of an increase in consumer confidence on GDP is therefore *lower*. **The statement is TRUE.**

MODEL ANSWER — WRITE IT LIKE THIS

True. An increase in consumer confidence raises consumption at every price level and shifts aggregate demand to the right. The size of the resulting increase in GDP depends on the slope of the short-run aggregate supply curve. The SRAS curve slopes upward because higher output raises employment and lowers unemployment, which strengthens workers' bargaining power and raises bargained nominal wages, which firms pass on into higher prices. If nominal wages are *highly* sensitive to unemployment, then even a small expansion of output triggers a large rise in wages and therefore in prices: the SRAS curve is steep. With a steep SRAS, a given rightward shift of AD translates mostly into a higher price level and only slightly into higher output. Hence the higher the sensitivity of bargained wages to unemployment, the smaller the short-run effect of the confidence shock on GDP. In the limiting case of extreme sensitivity the SRAS curve is vertical and the effect on GDP is zero, as in the classical long-run model.

Exam tip — the general principle behind Statement 2

Learn the general result, because it can be asked in many disguises: **the flatter the SRAS curve, the more a demand shock affects output and the less it affects prices; the steeper the SRAS curve, the more it affects prices and the less it affects output.** The same logic applies to the AD slope: *the flatter the AD curve, the more a supply shock affects output rather than prices.* Any question of the form "the more X, the smaller the effect of a demand shock on GDP" is asking you to show that X makes SRAS steeper. Things that make SRAS steeper: wages more responsive to unemployment, contracts renegotiated more frequently, expectations that adjust faster, prices that adjust faster (fewer menu costs), an economy operating close to full capacity.

§14 Problems and applications — a method

SLIDE 36 — ON THE SLIDE

Problems and applications

Class exercises: 3 (b) · 7 · 9 (a, b) · 10 (a, b, c)

Other interesting exercises: 1 · 3 (a, c) · 4 · 5 (b, d) · 8 · 9 (c)

These numbers refer to the end-of-chapter "Problems and Applications" in Mankiw & Taylor, *Economics* (6th edition), Chapter 12 — work them from the book itself. What this guide can give you is the **protocol** that solves essentially all of them, plus the checks that catch the usual mistakes.

The five-step protocol

Step	Question to ask	How to answer it
1	Where do we start?	Unless told otherwise, assume a long-run equilibrium: draw LRAS vertical, AD crossing it, and SRAS through the same point. Label it A. Say in words: " $Y = Y_n$, $u = u_n$, $P^e = P$."
2	Which curve moves, and why?	Apply the triage from §9: does the event change C , I , G , NX at a given price level (AD); does it change costs or expectations (SRAS); does it change productive capacity or u_n (SRAS and LRAS)? Name the component or the cost that changes — that is where the marks are.
3	What is the short-run effect?	Move only that curve. Read off the new intersection: what happens to P , to Y , and (via Fact 3 of §1) to unemployment. Name the point B.
4	What is the long-run effect?	Compare actual P at B with the expected P that SRAS was drawn for. If $P > P^e$, expectations and wages rise and SRAS shifts left/up ; if $P < P^e$, they fall and SRAS shifts right/down . Continue until the economy is back on LRAS. Name the point C. State what has changed <i>permanently</i> and what has not.
5	What could policy do?	Monetary and fiscal policy move AD only . Say which direction, what it achieves, and what it costs. For a supply shock, always mention that AD policy cannot fix output and prices simultaneously.

The three sanity checks

- **Check the long-run output.** Unless the event changed labour, capital, natural resources, technology or u_n , long-run output **must** return to the original Y_n . If your answer has a demand shock permanently changing output, you have made an error.
- **Check the direction of the SRAS adjustment.** It always moves so as to *close* the gap between actual and expected prices — i.e. so as to push output back towards Y_n . If your

SRAS shift takes output further away from Y_n , the sign is wrong.

- **Check the hidden interest rate.** If the question mentions investment, the interest rate, or the money market, add the IS-LM layer: a higher price level always means lower M/P , LM left, higher i ; a lower price level means the opposite.

Exam tip — what a full-mark diagram contains

Axes labelled **Price level** and **Quantity of output**; a vertical LRAS labelled at the **natural rate of output**; AD and SRAS labelled with subscripts (AD_1 , AD_2 , AS_1 , AS_2); the initial equilibrium marked A and every subsequent equilibrium marked B, C...; arrows showing the direction each curve moves; and the relevant P and Y values marked on the axes with dotted lines. Draw it big — a cramped diagram is a diagram in which the examiner cannot see whether B is left of the LRAS line.

A One-page cheat sheet

The three curves

Curve	Shape	Why that shape	What shifts it
AD	Downward sloping	Wealth effect ($\rightarrow C$): lower P raises the real value of nominal assets. Interest-rate effect ($\rightarrow I$): lower P raises M/P, LM right, i falls, I rises. <i>Main channel; derived from IS-LM.</i> Exchange-rate effect ($\rightarrow NX$): lower i depreciates the currency, NX rises.	Anything changing C , I , G or NX at a given P : consumer confidence, income taxes and transfers, business confidence, monetary policy (M) , fiscal policy (G), foreign income, the exchange rate.
SRAS	Upward sloping	Sticky wage theory: nominal wages are slow to adjust because price expectations adapt slowly and contracts are revised infrequently. Chain: demand $\uparrow \rightarrow$ production $\uparrow \rightarrow$ employment $\uparrow \rightarrow$ unemployment $\downarrow \rightarrow$ bargained wages $\uparrow \rightarrow$ prices set by firms $\uparrow$. (Bracketed: sticky price / menu costs; misperceptions.)	Expected price level P^e (SRAS only), plus the four LRAS shifters: labour (incl. u_n), capital, natural resources (e.g. an <i>oil-price shock</i>), technology.
LRAS	Vertical at Y_n	Output in the long run depends on labour, capital, natural resources and technology — none of which depend on the price level. Monetary neutrality.	Labour, capital, natural resources, technological knowledge — i.e. anything changing productive capacity or the natural rate of unemployment. Never demand-side events.

The four shocks — short-run and long-run outcomes (starting from long-run equilibrium)

Shock	SR Y	SR P	SR u	Long run, once expectations adjust
AD increases (confidence $\uparrow$, $G \uparrow$, $M \uparrow$, taxes $\downarrow$, foreign income $\uparrow$)	$\uparrow$	$\uparrow$	$\downarrow$	P^e and wages rise $\rightarrow$ SRAS left $\rightarrow$ back to Y_n with a permanently higher price level .
AD decreases (Figure 7)	$\downarrow$	$\downarrow$	$\uparrow$	P^e and wages fall $\rightarrow$ SRAS right $\rightarrow$ back to Y_n with a permanently lower price level .
Adverse SRAS shift (oil shock — Figure 8; <i>stagflation</i>)	$\downarrow$	$\uparrow$	$\uparrow$	If temporary: high unemployment lowers P^e and wages $\rightarrow$ SRAS back right $\rightarrow$ return to the original Y and P . Or policy accommodates $\rightarrow Y_n$ restored at a higher price level (Figure 9).
Favourable LRAS shift (technology, capital, labour supply)	$\uparrow$	$\downarrow$	$\downarrow$	Both SRAS and LRAS shift right: permanently higher output and a lower price level. This is the growth trend of Figure 4.

Movement versus shift — the discriminating question

Event	Movement or shift?	Reason
The price level falls	Movement along AD (and along SRAS)	P is the variable on the axis.
The central bank raises M	Shift of AD (right)	M/P changes through the <i>numerator</i> .
The price level rises, so M/P falls	Movement along AD	M/P changes through the <i>denominator</i> .
Unions revise inflation expectations up	Shift of SRAS (left/up)	P^e is a shift parameter of SRAS; LRAS unaffected.
Oil prices jump	Shift of SRAS (left/up)	Costs rise at every output level.
Immigration raises the labour force	Shift of SRAS and LRAS (right)	Productive capacity rises.

The eight sentences worth memorising verbatim

1. Economic fluctuations are irregular and unpredictable; most macroeconomic variables fluctuate together, though by different amounts; and as output falls, unemployment rises.
2. Classical theory describes the long run but not the short run: monetary neutrality is not appropriate for year-to-year changes.
3. The AD curve is derived from the IS-LM model; the AS curve is derived from the wage- and price-setting behaviour of workers and firms.

4. A higher price level reduces real money balances, raises the interest rate, and reduces investment and aggregate demand.
5. The LRAS curve is vertical at the natural rate of output, the level of production for which the unemployment rate equals the natural rate of unemployment.
6. Nominal wages are sticky because price expectations adapt slowly and contracts are revised infrequently.
7. Long-run equilibrium is restored once expected prices equal actual prices.
8. In the short run, shifts in aggregate demand cause fluctuations in output; in the long run, they affect the price level but not output.

B Glossary of examinable terms

Term	Definition
Business cycle	The irregular and largely unpredictable fluctuations in economic activity — output, incomes and employment — around the economy's long-run trend.
Recession	A period of falling real income and rising unemployment; commonly identified with two consecutive quarters of falling real GDP.
Depression	A severe and prolonged recession.
Classical dichotomy	The theoretical separation of economic variables into real variables (quantities and relative prices) and nominal variables (measured in money units).
Monetary neutrality	The proposition that changes in the money supply affect nominal variables but not real variables. Holds in the long run; not in the short run.
Aggregate demand curve	A curve showing the quantity of goods and services that households, firms and the government want to buy at each price level. Derived from the IS-LM model.
Aggregate supply curve	A curve showing the quantity of goods and services that firms choose to produce and sell at each price level. Derived from the wage- and price-setting behaviour of workers and firms.
Wealth effect	A lower price level raises the real value of households' money holdings and other nominally fixed assets, making consumers feel wealthier, so consumption rises.
Interest-rate effect	A higher price level reduces real money balances M/P ; the interest rate on the money market rises; investment falls; aggregate demand falls.
Exchange-rate effect	A higher price level raises the interest rate, attracting capital inflows and appreciating the currency, which reduces net exports.
Real money balances	M/P — the purchasing power of the money supply; the quantity of goods and services the money stock can buy.
Real wage	W/P — the amount of goods and services that can be bought with a nominal wage.
Adaptive expectations	Expectations of the future price level formed on the basis of past inflation, so they adjust only gradually to changes in actual inflation.
Natural rate of output (Y_n)	The output an economy produces when unemployment is at its natural rate; also potential output or full-employment output.
Natural rate of unemployment (u_n)	The unemployment rate towards which the economy gravitates in the long run — frictional plus structural unemployment. Not zero.
Sticky wage theory	Nominal wages adjust slowly because price expectations adapt slowly and wage contracts are revised infrequently; this makes the SRAS curve upward sloping.
Sticky price theory	Some prices adjust slowly because of menu costs, so an unexpected fall in the price level leaves some firms with prices too high, cutting their sales and their production.
Misperceptions theory	Producers confuse changes in the overall price level with changes in the relative price of their own product and adjust output accordingly.

Long-run equilibrium	The situation in which expected prices equal actual prices, so there is no pressure to revise wages or prices: AD, SRAS and LRAS intersect at a single point, with $Y = Y_n$ and $u = u_n$.
Stagflation	A period of recession (falling output, rising unemployment) combined with inflation (a rising price level), caused by an adverse shift in aggregate supply.
Accommodating a supply shock	Expanding aggregate demand in response to an adverse supply shock, restoring output to its natural rate at the cost of a permanently higher price level.
Okun's law	The empirical inverse relationship between changes in real GDP and changes in the unemployment rate.
Labour hoarding	Firms retaining trained workers during a downturn rather than dismissing them, because hiring and training are costly; it makes employment fall by less than output.

C The fifteen classic traps

#	The mistake	The correction
1	Explaining the AD slope with a substitution argument ("goods are cheaper so people buy more").	Along AD <i>all</i> prices move together — there is nothing to substitute towards. Use the wealth, interest-rate and exchange-rate effects.
2	Confusing the price level with the inflation rate .	The vertical axis is the level of prices (an index). Inflation is its rate of change. A one-off rightward AD shift raises the price <i>level</i> permanently; it does not create permanent inflation.
3	Treating a change in M and a change in P as the same thing because both change M/P .	Numerator (M) $\Rightarrow$ AD shifts . Denominator (P) $\Rightarrow$ movement along AD.
4	Shifting AD and SRAS simultaneously when a single shock hits.	Move only the shocked curve first, read the short run, and only then let SRAS respond to the forecast error.
5	Getting the SRAS adjustment backwards after a demand contraction.	Actual P below expected $P^e \Rightarrow$ expectations and wages fall $\Rightarrow$ SRAS shifts right/down .
6	Letting a demand shock change output permanently.	Unless capacity or u_n changed, long-run output must return to Y_n . Only the price level ends up different.
7	Thinking "full-employment output" means zero unemployment.	It means $u = u_n$ — frictional and structural unemployment remain.
8	Believing output can never exceed Y_n .	It can, temporarily, in a boom (overtime, higher utilisation, u below u_n). It is unsustainable, not impossible.
9	Shifting LRAS because unemployment rose in a recession.	Cyclical unemployment is a movement <i>away from</i> LRAS. Only a change in u_n or in capacity moves the vertical line.
10	Shifting LRAS when the expected price level changes.	P^e is nominal: it shifts SRAS only.
11	Claiming AD policy can fully offset a supply shock.	It cannot. AD policy moves P and Y in the same direction; a supply shock moves them in opposite directions. You must choose which problem to treat.
12	Forgetting the interest rate when asked about investment.	The AD curve hides an IS-LM diagram. Higher $P \Rightarrow$ lower $M/P \Rightarrow$ LM left $\Rightarrow$ higher $i \Rightarrow$ lower I . This is the trick in Statement 1.
13	Saying "demand goes up" when a curve shifts.	Say "at each price level, the quantity demanded is higher". Reserve "quantity demanded" for movements along the curve.
14	Drawing SRAS crossing LRAS at a different point from AD in a long-run equilibrium.	All three must pass through one point. Draw LRAS first, then AD, then SRAS through the intersection.
15	Treating the wealth effect as the main reason AD slopes down.	The interest-rate effect is the main channel in this course — it is the one the professor derives with the IS-LM diagram. The wealth effect is real but quantitatively modest.

D Practice multiple-choice questions

Answers with explanations follow the questions. Cover them and work through the set in one sitting; anything you get wrong points to the section you should re-read.

1. Which of the following is *not* one of the three key facts about economic fluctuations?
 - A. Fluctuations are irregular and unpredictable.
 - B. Most macroeconomic variables fluctuate together.
 - C. As output falls, unemployment rises.
 - D. Fluctuations follow a regular cycle of roughly seven years.
2. Which component of GDP does *not* respond to a change in the price level along the AD curve?
 - A. Consumption
 - B. Investment
 - C. Government purchases
 - D. Net exports
3. According to the interest-rate effect, an increase in the price level:
 - A. reduces real money balances, raises the interest rate and reduces investment.
 - B. raises real money balances, lowers the interest rate and raises investment.
 - C. raises the real value of money holdings and increases consumption.
 - D. depreciates the currency and raises net exports.
4. The central bank increases the money supply while the price level is unchanged. In the AD-AS diagram this:
 - A. is a movement along the AD curve.
 - B. shifts the AD curve to the right.
 - C. shifts the SRAS curve to the right.
 - D. shifts the LRAS curve to the right.
5. The long-run aggregate supply curve is vertical because:
 - A. nominal wages are perfectly sticky.
 - B. in the long run, output depends on labour, capital, natural resources and technology, none of which depend on the price level.
 - C. the central bank targets output.
 - D. aggregate demand is vertical in the long run.
6. Which of the following shifts *both* SRAS and LRAS to the right?
 - A. A fall in the expected price level.
 - B. A permanent improvement in production technology.
 - C. An increase in the money supply.
 - D. A rise in consumer confidence.

7. An increase in the expected price level:

- A. shifts SRAS to the left and LRAS to the left.
- B. shifts SRAS to the left and leaves LRAS unchanged.
- C. shifts AD to the left.
- D. has no effect on any curve.

8. In the short run, an adverse shift in aggregate supply causes:

- A. lower output and a lower price level.
- B. lower output and a higher price level.
- C. higher output and a higher price level.
- D. higher output and a lower price level.

9. Starting from long-run equilibrium, aggregate demand falls. In the long run, the economy returns to the natural rate of output because:

- A. aggregate demand automatically shifts back to the right.
- B. the LRAS curve shifts to the left.
- C. expected prices and bargained nominal wages fall, shifting SRAS to the right.
- D. the government is obliged to intervene.

10. The natural rate of output is best described as:

- A. the maximum output the economy could ever produce.
- B. the output at which unemployment is zero.
- C. the output at which the unemployment rate equals the natural rate of unemployment.
- D. the output level that maximises the price level.

11. Which condition characterises long-run equilibrium in the AD-AS model?

- A. The expected price level equals the actual price level.
- B. The money supply is constant.
- C. Net exports are zero.
- D. The interest rate is zero.

12. If the SRAS curve is relatively flat, a rightward shift in aggregate demand will:

- A. raise output a lot and the price level only a little.
- B. raise the price level a lot and output only a little.
- C. raise the price level without changing output.
- D. have no effect in the short run.

13. According to the sticky wage theory, nominal wages adjust slowly because:

- A. workers care only about nominal wages.
- B. price expectations adapt slowly and wage contracts are revised infrequently.
- C. firms cannot observe the price level.
- D. the government fixes wages.

14. Which of the following is a *movement along* the aggregate demand curve rather than a shift of it?

- A. A cut in income taxes.
- B. An increase in foreign income.
- C. A fall in the overall price level, other things equal.
- D. A collapse in business confidence.

15. An economy sits at a point to the left of its LRAS curve, with the actual price level below the expected price level. Over time:

- A. SRAS shifts right, output rises and the price level falls further.
- B. SRAS shifts left, output falls further and the price level rises.
- C. LRAS shifts left to meet the economy.
- D. nothing happens without policy intervention.

16. "Accommodating" an adverse supply shock means:

- A. contracting aggregate demand to bring the price level back down.
- B. expanding aggregate demand so output returns to the natural rate at a higher price level.
- C. waiting for wages and expectations to adjust.
- D. shifting the LRAS curve to the right.

17. In the long run, a permanent increase in the money supply raises:

- A. output only.
- B. the price level only.
- C. both output and the price level.
- D. neither.

18. Investment is the most volatile component of GDP over the business cycle mainly because:

- A. it is the largest component of GDP.
- B. it is postponable and highly sensitive to interest rates and expectations.
- C. it is set by the government.
- D. it does not depend on the price level.

19. In this course, the aggregate demand curve is derived from:

- A. the aggregate production function.
- B. the IS-LM model.
- C. the wage- and price-setting behaviour of workers and firms.
- D. the Phillips curve.

20. Compared with an adverse demand shock, an adverse supply shock (with the money supply unchanged) leads in the short run to:

- A. a lower interest rate, because output falls.
- B. a higher interest rate, because the price level rises and real money balances fall.
- C. an unchanged interest rate, because the money supply is unchanged.
- D. a lower price level.

21. If bargained nominal wages become *more* sensitive to the unemployment rate, the SRAS curve becomes:

- A. flatter.
- B. steeper.
- C. horizontal.
- D. unchanged in slope but shifted right.

22. A rise in the natural rate of unemployment caused by an institutional reform of the labour market:

- A. shifts LRAS left and SRAS left.
- B. shifts LRAS left only.
- C. shifts AD left.
- D. has no effect on any curve.

23. Figure 4 ("Long-run growth and inflation") shows that, over decades:

- A. output growth comes from rightward shifts of LRAS and inflation from rightward shifts of AD.
- B. output growth comes from rightward shifts of AD and inflation from shifts of LRAS.
- C. output and prices are both determined by the money supply.
- D. the economy is permanently off its LRAS curve.

24. Moving upward along a given aggregate demand curve, the interest rate:

- A. rises, because the higher price level lowers real money balances.
- B. falls, because output is lower.
- C. is unchanged, because the money supply is unchanged.
- D. cannot be determined.

25. In the long run, an increase in aggregate demand affects:

- A. output only.
- B. the price level only.
- C. both output and the price level.
- D. neither output nor the price level.

- 26.** Which policy response would restore *both* the original output level *and* the original price level after a fall in aggregate demand?
- A. Expansionary monetary or fiscal policy that shifts AD back to the right.
 - B. Contractionary monetary policy.
 - C. A policy that shifts SRAS to the left.
 - D. No policy can restore both.

Answer key with explanations

Q	Ans	Why
1	D	The first key fact is precisely that fluctuations are <i>irregular</i> — the word "cycle" is misleading because there is no fixed period. §1.
2	C	G is set by political decision and does not respond to the price level. C responds via the wealth effect, I via the interest-rate effect, NX via the exchange-rate effect. §4.
3	A	The four links from slide 11: $P \uparrow \rightarrow M/P \downarrow \rightarrow i \uparrow \rightarrow I \downarrow \rightarrow AD \downarrow$. Option C is the wealth effect stated in the wrong direction; D is the exchange-rate effect in the wrong direction. §4.
4	B	M/P changes through the numerator, so LM shifts for a reason other than P : the whole AD curve moves. Compare with Q24. §5.
5	B	The price level does not appear in the production function; this is monetary neutrality on the supply side. §6.
6	B	Technology changes productive capacity, so both supply curves move. A shifts SRAS only; C and D shift AD only. §9.
7	B	Higher $P^e \rightarrow$ higher bargained wages $\rightarrow$ higher costs $\rightarrow$ SRAS left. LRAS depends only on real factors. §8–§9.
8	B	The economy moves up along an unchanged AD curve: output falls, the price level rises — stagflation. Figure 8, §12.
9	C	Actual prices come in below expectations and unemployment is high, so expectations and wages fall, costs fall and SRAS shifts right (B $\rightarrow$ C in Figure 7). §11.
10	C	Exactly the wording of slide 18. It is not a maximum, and it is not zero unemployment. §6.
11	A	Slide 26: long-run equilibrium is restored once ex-ante expected prices equal ex-post actual prices. §8, §10.
12	A	A flat SRAS means prices respond weakly to output, so the demand shift is absorbed mostly by quantities. This is the general principle behind Statement 2. §13.
13	B	The two reasons on slide 23. Option A contradicts slide 22, which says workers care about the <i>real</i> wage. §7.
14	C	Only a change in the price level itself produces a movement along AD; the others all change spending at a given price level. §5.
15	A	The forecast error is negative, so expectations and wages are revised down and SRAS shifts right; the economy slides down along AD, output rising and prices falling further. §8, §11.
16	B	Figure 9: AD is expanded so output returns to Y_n , at the cost of $P_3 > P_2 > P_1$. §12.
17	B	Money is neutral in the long run: output returns to Y_n and only the price level is permanently higher. §2, §11.
18	B	Investment is postponable, interest-sensitive and expectations-driven; it is <i>not</i> the largest component of GDP. §1.
19	B	Slide 5 says so explicitly; option C is the derivation of the <i>supply</i> curve. §3.

20	B	A supply shock raises P , so M/P falls, LM shifts left and i rises — the opposite of a demand-driven recession. This is the key to Statement 1. §12–§13.
21	B	Greater wage responsiveness means a small rise in output triggers a large rise in wages and hence prices: SRAS is steeper. §13.
22	A	A higher u_n lowers the natural rate of output, so both supply curves shift left. Slide 27 lists this explicitly under "Labour". §9.
23	A	The supply side delivers real growth (technology, capital, labour); money growth shifts AD right and delivers the inflation. §6.
24	A	Moving up along AD means a higher P with M unchanged, so M/P falls, LM shifts left and i rises. The money supply being constant is exactly <i>why</i> the interest rate moves. §4.
25	B	Slide 29, second bullet: in the long run, shifts in AD affect the overall price level but not output. §11.
26	A	For a <i>demand</i> shock, reversing the shift restores both variables simultaneously — the two problems point in the same direction. (Contrast with a supply shock, where no AD policy can do both.) §11–§12.

E Practice open questions with model answers

Question 1. Explain why the aggregate demand curve slopes downward. Which of the effects you mention is the most important in this course, and why?

MODEL ANSWER

The AD curve slopes downward because a lower price level raises the total quantity of goods and services demanded, through three channels, each working on a different component of $Y = C + I + G + NX$.

The wealth effect (consumption). Households hold part of their wealth in assets with a fixed nominal value, chiefly money. A lower price level raises the real value of these holdings, so households feel wealthier and increase consumption.

The interest-rate effect (investment). The central bank fixes the nominal money supply M . A lower price level raises real money balances M/P . In IS-LM terms the LM curve shifts to the right; on the money market, the excess supply of real balances drives the interest rate down. A lower interest rate makes more investment projects profitable, so investment rises and aggregate demand rises.

The exchange-rate effect (net exports). The lower domestic interest rate induces capital outflows, the currency depreciates, domestic goods become more competitive and net exports rise.

Government purchases do not respond to the price level, so there is no fourth channel.

The **interest-rate effect is the most important** in this course: it is the channel through which the AD curve is formally derived from the IS-LM model (each point on AD is an IS-LM equilibrium for a given price level), and it is quantitatively the largest. The wealth effect is limited because money is a small share of wealth and because the gains of creditors are largely offset by the losses of debtors; the exchange-rate effect is treated as secondary.

Question 2. Explain why the short-run aggregate supply curve slopes upward, using the wage- and price-setting behaviour of workers and firms.

MODEL ANSWER

Nominal wages are bargained between unions and employers, and they depend on two things: the **expected price level** and the **unemployment rate**. Workers care about the real wage W/P — what their wage can buy — but the price level over the coming contract period is unknown when the bargain is struck, so they must use an expectation P^e , which is formed adaptively on the basis of past inflation. A higher unemployment rate weakens workers' bargaining power and therefore lowers the bargained nominal wage. Firms then set their prices as a mark-up over their costs, of which wages are the largest part.

Now consider an increase in demand. Firms must produce more, so employment rises and unemployment falls. Lower unemployment strengthens workers' bargaining position, so bargained nominal wages rise. Firms' unit costs rise, and they set higher prices to cover them. Higher output is therefore associated with a higher price level: the short-run aggregate supply curve slopes upward.

The relationship holds only in the **short run** because nominal wages are sticky: expected prices adapt only slowly to actual prices, and wage contracts are revised infrequently. The higher prices set by firms do not immediately feed back into higher bargained wages. Once they do — once expectations catch up — the SRAS curve shifts and the extra output disappears.

Question 3. The economy starts in long-run equilibrium. Consumer confidence collapses. Describe the short-run and long-run effects on output, the price level and unemployment, and explain the adjustment mechanism. What could policymakers do?

MODEL ANSWER

The shock. A collapse in consumer confidence reduces consumption at every price level, so aggregate demand shifts to the left, from AD_1 to AD_2 .

Short run (A → B). Nominal wages are fixed by contracts and expectations have not been revised, so SRAS does not move. The economy slides down along the unchanged SRAS curve. Output falls below the natural rate, the price level falls, and unemployment rises above its natural rate. The fall in prices is only partial, which is precisely what an upward-sloping SRAS means.

Adjustment (B → C). At B, the actual price level is *below* the level wage setters had expected, and unemployment is high. Both push wages down: expectations of prices are revised downward, and high unemployment weakens workers' bargaining power. Lower bargained nominal wages reduce firms' unit costs, so at any given price level firms are willing to supply more — the SRAS curve shifts to the right. The economy slides down along AD_2 .

Long run (C). The process stops when expected prices again equal actual prices. Output is back at the natural rate and unemployment back at its natural rate, but the price level is permanently lower. The demand shock has had a *temporary* effect on output and a *permanent* effect on the price level — the classical result of monetary neutrality, recovered in the long run.

Policy. Self-correction works, but slowly, and the whole adjustment period is one of elevated unemployment. Policymakers can instead shift AD back to the right by expansionary monetary policy (increasing the money supply, which lowers the interest rate and raises investment) or expansionary fiscal policy (higher government purchases, lower taxes or higher transfers). Because the shock and the remedy both operate on aggregate demand, this restores *both* output and the original price level. The counter-argument is that policy operates with lags and may destabilise rather than stabilise.

Question 4. What is stagflation, what causes it, and why is it particularly difficult for policymakers?

MODEL ANSWER

Stagflation is the combination of a recession — falling output and rising unemployment — with a rising price level. It is caused by an **adverse shift in short-run aggregate supply**: an event that raises firms' costs of production at every level of output, such as a sharp increase in oil prices following a decision of the OPEC cartel. Firms are willing to supply less at each price level (equivalently, they must charge more to produce each level of output), so SRAS shifts left/upward. Since aggregate demand has not moved, the economy travels up along the AD curve: output falls below the natural rate, unemployment rises, and the price level rises.

The difficulty is that monetary and fiscal policy can influence only **aggregate demand**, and shifting AD always moves output and the price level in the *same* direction. The supply shock has moved them in *opposite* directions. Therefore policymakers cannot offset both adverse effects at once:

- *Accommodate* — expand AD: output returns to its natural rate, but the price level rises further still.
- *Fight the inflation* — contract AD: the price level is pushed back down, but the recession deepens.
- *Do nothing* — wait: high unemployment eventually lowers price expectations and bargained wages, SRAS shifts back to the right, and the economy returns to the original output and price level. This avoids permanent inflation but at the cost of a prolonged recession.

A further consideration against accommodation is that validating the price increase can entrench higher inflation expectations, which shifts SRAS left again and can set off a wage-price spiral.

Question 5. Explain the difference between a movement along the aggregate demand curve and a shift of the aggregate demand curve, giving one example of each that involves the money market.

MODEL ANSWER

A **movement along** the AD curve is caused by a change in the price level and nothing else; it is a change in the *quantity of goods and services demanded*. A **shift** of the AD curve is caused by a change in any other determinant of C , I , G or NX at a given price level; it is a change in *aggregate demand* itself.

Both examples involve real money balances M/P , which is what makes this distinction tricky:

- **Movement along.** The price level rises while the central bank holds M constant. Real money balances fall because the *denominator* rose. The LM curve shifts left, the interest rate rises, investment falls, and the quantity of output demanded falls. The economy moves up and to the left *along the same AD curve* — indeed this is exactly how the AD curve is derived.
- **Shift.** The central bank increases the money supply while the price level is unchanged. Real money balances rise because the *numerator* rose. The LM curve shifts right, the interest rate falls, investment rises — but at an unchanged price level. Since the quantity demanded is now higher at *every* price level, the whole AD curve shifts to the right.

The discriminating question is therefore always: *did M/P change because of the numerator or the denominator?*

Question 6. Why is the long-run aggregate supply curve vertical, and what could shift it? Explain why a recession does not shift it.

MODEL ANSWER

In the long run, an economy's production of goods and services depends on its supplies of labour, capital and natural resources and on the technology available to combine them. None of these depends on the price level: if all prices and all wages doubled, the same workers with the same machines and the same knowledge could produce exactly the same real output. The quantity supplied is therefore the same at any price level, and the LRAS curve is a vertical line at the natural rate of output — the level of production at which unemployment equals its natural rate. This is monetary neutrality expressed on the supply side.

The LRAS curve shifts whenever something alters the natural rate of output. Following the classical production function, the shifters are: **labour** (population growth, migration, participation, and changes in the natural rate of unemployment); **capital** (physical and human capital accumulation); **natural resources** (discoveries or permanent losses of access); and **technological knowledge** (innovation, and also trade liberalisation, which acts like a technological improvement).

A recession does *not* shift LRAS. In a recession, unemployment rises above its natural rate and output falls below potential, but nothing has happened to the labour force, the capital stock, natural resources, technology, or the natural rate of unemployment. The economy has simply moved *away from* its LRAS curve; the curve itself has not moved. That is precisely why the economy eventually returns to it.

Question 7. "In the short run, shifts in aggregate demand cause fluctuations in output; in the long run, they affect only the price level." Explain the economics behind this sentence.

MODEL ANSWER

The sentence follows from the difference in the shape of the aggregate supply curve over the two horizons.

Short run. Nominal wages are sticky, because price expectations adapt slowly and contracts are renegotiated infrequently. As a result, the SRAS curve is *upward sloping*: firms respond to a change in demand partly by changing prices and partly by changing production. When AD shifts, the new intersection therefore lies at a different output level as well as a different price level. Output fluctuates, and unemployment fluctuates with it in the opposite direction.

Long run. Eventually wage setters correct their forecast error. If the actual price level exceeded expectations, expectations are revised up, bargained nominal wages rise, firms' costs rise and SRAS shifts left; if it fell short, the reverse happens and SRAS shifts right. The shifting continues until expected prices equal actual prices, which is exactly the condition that puts the economy back on the vertical LRAS curve. Since LRAS is vertical at the natural rate of output — determined by labour, capital, natural resources and technology, none of which the demand shock touched — output must return to where it started. The only variable that can absorb the shift permanently is the price level.

In other words, monetary neutrality fails in the short run because of nominal rigidities, and is restored in the long run once those rigidities have worked themselves out.

Question 8. Two economies are identical except that in country A wage contracts are renegotiated every three years, while in country B they are renegotiated every six months. Both experience the same increase in government purchases. Compare the short-run effects on output and prices, and the speed of adjustment to the long run.

MODEL ANSWER

The increase in government purchases shifts AD to the right in both countries by the same amount. The difference lies in the slope of the SRAS curve and in the speed at which it subsequently shifts.

Short-run effects. In country A, with three-year contracts, nominal wages are highly sticky: an expansion of output and the associated fall in unemployment translate only weakly and slowly into higher bargained wages, so firms' costs and hence prices rise little. The SRAS curve is relatively **flat**, and the demand expansion produces a **large increase in output and a small increase in the price level**. In country B, with six-month contracts, wages respond quickly to labour-market conditions, so a small expansion of output already triggers a noticeable rise in wages and prices. The SRAS curve is relatively **steep**, and the same demand expansion produces a **smaller increase in output and a larger increase in the price level**.

Speed of adjustment. In both countries the actual price level now exceeds what wage setters expected, so expectations are revised upward, wages rise, costs rise and SRAS shifts left until output returns to the natural rate. Country B completes this adjustment **much faster**, because contracts come up for renewal twice a year. Country A remains above its natural rate of output — with unemployment below its natural rate — for considerably longer.

Long run. Both countries end up at the same place: output back at the natural rate and a permanently higher price level. Nominal rigidity affects how large and how long-lasting the real effects of a demand shock are; it does not affect the long-run outcome. Country B behaves more like the classical model, and in the limiting case of instantaneous wage adjustment its SRAS curve would be vertical and the fiscal expansion would raise prices without ever raising output.

FINAL WORD BEFORE THE EXAM

If you remember nothing else on the day: **AD comes from IS-LM, AS comes from wage and price setting. AD shifts move output in the short run and prices in the long run. AS shifts move output and prices in opposite directions, which is why supply shocks are the hard ones. And everything returns to the natural rate of output when expected prices catch up with actual prices.** Every question in this chapter is a variation on those four sentences.